Geography in Reduced Form

by

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Abstract

Geography models have introduced and estimated a set of competing explanations for the persistent relationships between firm and location characteristics, but cannot identify these forces. I introduce a solution method for models in arbitrary geographies that generates reduced-form predictions and tests to identify forces acting through geographic linkages. This theoretical approach creates a new strategy for spatial empirics. Using the correct observables, the model shows that geographic forces can be taken into account without being directly estimated; establishment and employment density emerge as sufficient statistics for all geographic forces. I present two applications. First, the model can be used to evaluate whether geographic linkages matter and when simplified models suffice: the mono-centric model is a good fit for business services firms but cannot capture the geography of manufactures. Second, the model generates reduced-form tests that distinguish between spillovers and firm sorting and finds evidence of sorting.

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1. Introduction

The quantity and quality of economic activity varies dramatically across space. Larger, more productive firms are clustered in larger, denser cities.[1] Economists have sought to explain these consistent patterns by hypothesizing forces such as transportation costs and other pecuniary and non-pecuniary spillovers that drive the spatial differences in density and the ubiquitous relationships between firm and location characteristics.[2] Spatial models account for these forces by embedding them in models in which firms and workers choose from a network of interconnected locations with arbitrary geographic linkages. These models conclude that geographic linkages are important.[3]

However, technical limitations resulting from modeling geographic interconnections prevent these models from generating reduced-form tests for the presence of these forces. Such tests are only possible in urban economic models,[4] which generate reduced-form relationships in settings with restricted or no geographic linkages between locations.[5] But because such models abstract from the very pathways through which these geographic

[1] Ciccone & Hall (1996) were the first to differentiate between density and city size with respect to productivity. Recently work by Maré et al. (2006) and Combes et al. (2010) confirm the relationship.

[2] For an overview of such models in urban economics, seeRosenthal & Strange (2004a, 2001).

[3] A recent set of papers have developed methods to structurally estimate these forces in real urban spaces. See Allen & Arkolakis (2013) and Ahlfelt et al. (2012) for recent examples.

[4] In general, new geography models which can be estimated in arbitrary geographies can only yield analytic solutions when returning to restricted geographies. Allen & Arkolakis (2013), obtain closed-form solutions when applying the model to line segments and the model in the follow-up paper, Allen Arkolakis 2015, equilibrium conditions have analytic expressions on a circle-geography.

[5] The workhouse models of urban economics abstract from spatial networks. In the Rosen-Roback (1979; 1982) inter-city framework, the endogenous and exogenous characteristics of New York – but never its proximity to Boston and Philadelphia – dictate its economic outcomes. In the Alonso-Muth-Mills (1964; 1969; 1967) (AMM) intra-city framework, each location's proximity is modeled as distance to a central business district (CBD), and is always an exogenous characteristic. For a

forces operate, these tests can misidentify these forces and magnitudes.[6][7]

This paper presents a new solution method for geography models that generates testable, reduced-form predictions while explicitly modeling flexible geographic interconnections. In doing so, it offers a bridge between the recent developments in economic geography and urban economics and uncovers new evidence for the spatial forces described by both fields. I build a geography model where heterogeneous firms choose locations in an arbitrary geography with a broad set of geographic forces affecting those location decisions. I use the firms' location decision to construct an index that ranks locations according to the way these geographic forces affect firm revenues and solve the model using this index. This method circumvents the technical issues facing geography models – multiplicity and lack of analytic solutions – that prevent reduced-form analysis.

The model then yields a set of observable sufficient statistics for the unobserved index values: establishment density, employment density, and land prices. Using these variables as a proxy for the underlying geography-based index means that geographic forces can be taken into account without being directly (structurally) estimated; ordering locations according to establishment or employment density reveals the order of index values, which includes the combined value of all geographic forces including access to markets, inputs, and productivity spillovers from nearby firms. This delivers a set of predictions that relate firm and location characteristics which hold in any equilibrium and therefore characterize the full set of equilibria.

I present two applications of this new strategy for reduced-form spatial

[6]Tests based in the latter framework have found evidence of productivity spillovers and limited evidence for the importance of market access and firm sorting. See Rosenthal & Strange (2004b)for an overview of urban evidence on the sources of spatial agglomeration, and Greenstone et al. (2010) and Ellison et al. (2007) for classic examples of measurements of productivity spillovers and the relative strength of other agglomeration sources including market access, respectively.

[7]Monte et al. (2015) in particular find that estimates of the effects of place-based policies and local productivity shocks are significantly more variable than reduced-form estimates imply, and using local controls significantly bias estimates. They offer a correction based on their structural estimates that accounts for geographic linkages that specifically correct for bias resulting from commuting between U.S. counties.

empirics. The first demonstrates the empirical advantage of incorporating geographic linkages to reduced-form work. Accounting for geographic forces using density allows us to test the validity of traditional urban reduced-form empirics that lack geographic interconnections. I find the monocentric model works well as an approximation for the geography of business services firms but is unable to accurately capture that of manufactures. In the second application, I show how this approach can identify firm sorting and differentiate sorting from productivity spillovers. I find evidence that part of the density-productivity gradients in cities is due to sorting.

Section two introduces the model. The model nests the Allen and Arkolakis (2013) model. It is defined over a continuous space in with arbitrary geographic linkages, but allows for a wider set of geographic forces including geographically-based spillovers from nearby firms, and allows heterogeneous firms to choose locations. This framework can be considered a Rosen Roback(1979; 1982) model generalized to include geographic linkages and firm sorting between locations, as well as a bridge between the Rosen Roback model and the Alonso Muth Mills (1964; 1969; 1967) framework.[8]

In the model, a set of entrepreneurs that are differentiated by their productivities, choose locations for the production of differentiated goods that will be sold to all other agents in the space with trade costs. Larger firms attract more workers and, in equilibrium, differences in local productivity and access to markets arise endogenously. More productive firms outbid less productive firms for locations which endogenously become central, and landowners at those locations provide more floor space to more firms.

Section three solves for the equilibrium and introduces the solution method. I first construct the index,[9] which takes account of how the various geo-

[8]In an appendix section I show how the precise AMM framework can be recovered by assuming specific functional form for trade costs which restrict geographic linkages.

[9]The introduction of an index may resemble the approach of Davis and Dingel (2013), which also makes use of an index variable to solve a system of cities model. Davis and Dingel assign an exogenous amenity (that might be thought of as distance to CBD or something else) to all locations in each city. The amenity interacts with city size. In that they abstract from the problem of networked geography, Davis and Dingel (2013) is firmly within the traditional urban framework. By contrast, here, the index is endogenous and derived specif-

graphic forces affect revenues at each location. I show it is possible to express and solve all decisions with respect to the location index. In effect, this reduces a multi-dimensional problem into an assignment problem.

Section four shows how the model can be used to derive predictions and how establishment density, employment density, and land prices can be used as sufficient statistics for the unobserved index, and therefore for the full effect of all geographic forces on all other location and firm characteristics. These predictions hold in any equilibria.

The use of density as a sufficient statistic generates a check for the ability to test comparative statics under multiplicity. For a given density of index values and firms, predicted relationships between observables are unique. As a consequence, if in a given range, the number of tracts of each density and the average firm quality is consistent across time periods, comparative statics can be taken, even when the real geographic distribution of firms moves from one equilibrium to another across time periods.

The model generates density gradients, plots of firm characteristics against location density which are analogous to gradients generated with location distance in a monocentric city model. Section five uses establishment-level restricted-access census data to compare the density gradients predicted by the model to those of a monocentric model in which firms choose locations based on distance to a central business district (CBD) and firm qualities are predicted by the firm's distance to the CBD.

Density gradients are consistently predictive across all industries, while the monocentric model fails to accurately describe the geography of manufactures. The density gradients also confirm the hypothesis that density is a sufficient statistic for economic activity and reject the hypothesis that distance can be. These results can be viewed as an evaluation of the ability of the monocentric framework to capture geographic forces and suggest that it can be used without loss of significant informational for business service firms, but not for manufactures, which can only be geographically described

ically in order to take geographic linkages into account and simplify their expression as a point of departure from the urban framework.

in reduced form using the measures proposed here.

Section six introduces the second application, which uses the ordering of locations within cities to test for the existence of firm sorting. The model predicts that additional floor space in one location affects the productivity of entrants at other locations differentially based on the locations' relative ranks, with higher-ranked tracks experiencing increases in entrant productivity and lower-ranked tracks experiencing entrant productivity decreases. These differential effects are incompatible with any model in which there are no ex-ante differences in firm productivity.

I test for and find evidence of the existence of these composition effects. Doubling construction expenditures in a tract reduces the productivity of entrants at lower-ranked tracts by between 0.2% and 0.6%. Productivity of entrants at higher-ranked tracks increases by a similar percentage. The same exercise fails when tracts are ordered according to their distance to the CBD: the distance metric is too coarse a proxy to pick up on these fine, geographically-based predictions.

Because commercial real estate development requires liquid collateral, such firms are exposed to real estate shocks in multiple cities, and price shocks in one city can affect firm-level collateral and transfer resources away from projects in relatively lower-shocked cities. Instrumenting for the construction expenditures of a firm in one city using the sales of non-linked firms in linked cities, I find larger composition effects with similar differential signs.

This paper's main contribution is to a large and growing literature in economic geography which structurally estimates the effects of various geographic forces (Allen & Arkolakis, 2013; Ahlfelt et al., 2012). This model incorporates sorting and heterogeneity forces as well as a larger set of geographic forces into this class of models. Most importantly, it provides this literature with a method to derive analytic relationships from such models and bridges the results of this literature with the more established urban literature. The paper adds theoretically and empirically to a large literature on firm sorting (Behrens et al., 2010). The setting here replicates the

city-size productivity relationship in the urban theory literature as a density-productivity relationship, but here the relationship is an endogenous feature rather than assumed, and the geographic approach overturns two key results of that literature: in this setting, which accounts for between and within-city sorting, higher productivity cutoffs need not be observed in larger cities, and sorting and spillovers need not be complements.[10] The empirical test of sorting here provides the first reduced-form confirmation of the existing of firm sorting that is estimated by this literature. Finally, this paper lends guidance to an ambivalent empirical literature in urban economics for which the monocentric model continues to serve as the theoretical workhorse (Burchfield et al., 2006). The paper offers a way in which the empirical applicability of the monocentric or any other model can be measured and finds results that are largely dependent on industry.

2. Model Setup

In this section, I set out the optimization problems for each of the three agents: workers, landowners, and entrepreneurs. I then introduce the variable that will act as an index of a location's productive advantage.

The model can be thought of as a Melitz (2003) model where both a set of productivity-differentiated firms and a set of homogenous workers can choose their locations. Landowners at each location charge rents and provide floor space to firms, who sell their differentiated goods to consumers at all other locations and potentially capture spillovers from firms at their or other locations. The geographic linkages between locations and the location decisions of all agents interact to create endogenous differences in locations' access to consumers (markets) and other firms (spillovers).

Geography. The model is defined over S, any compact[11] subset of Eu-

[10]These results appear in Combes et al. (2012) and Gaubert (2014) respectively. Productivity cutoffs may be absent in the continuous-space model when large cities contain distance or disadvantageous locations which can only compete for low-productivity firms. In addition, when multiple geographic forces are present, sorting may occur on one dimension and not another. For example, when market access is important, firms may choose higher market access locations over locations with productivity spillovers, creating negative assortative matching between firm productivity and location productivity.

[11] The compact nature of the space guarantees the existence of a boundary, locations

clidean space R^n, $n \in \mathbb{N}$ with a non-finite set of points. This domain ensures the model can be applied to any realistic geography, especially one which might be constructed from data on countries, regions, or cities. All economic activities, production and consumption, take place at points $i \in S$.

Three kinds of agents take part in production and consumption in the space: landowners, workers, and entrepreneurs. The space is filled with a mass of immobile landowners, each endowed with a point $i \in S$, refer to as their location. Workers and entrepreneurs will choose locations (and landowners).

The space is further defined by a function governing transportation costs for goods and the geographic decay of spillovers between all points in the space, with continuous, differentiable, and symmetric cost function:

$$1 < \tau_l(i,j) < \infty,\ l \in 1,2,$$

from one point i to another point j. Goods are sold from i to j with iceberg transport costs $\tau_1(i,j)$ and spillovers between firms at those two locations will be mediated by a decay function defined by $\tau_2(i,j)$.

I impose symmetry and the triangle inequality on $\tau(i,j)$.[12] No other functional forms or assumptions need be placed on τ.

Locations differ by their relative proximity to other locations. On its own, this exogenous geography will not drive the location decisions of firms and

that are relatively distant. Given further assumptions on the transportation costs stated below, locations on the boundary will inevitably be economically remote, and thus less advantageous. This will be crucial in ensuring at least partial sorting.

[12]This is a technical assumption used to help satisfy the existence of a sorting equilibrium. It will be necessary in order to ensure that points on the boundary are actually more distant from all other locations than nearby points, so that market access and prices both improve away from the boundary. A less strict condition is sufficient. For any point k on the vector $\vec{ij}$, $\tau(i,j) > \tau(i,k)$ and $\tau(i,j) > \tau(k,j)$; intuitively for any journey between two points i and j, transportation costs are lower for stops along the way. If travel between i and j always takes place on the euclidean vector $\vec{ij}$, this constraint intuitively means shorter trips facing the same geography must be less costly. In a world where shortest cost trips may include circuitous routes, this constraint may be unrealistic.

workers. Rather, a location's advantage is determined by the equilibrium decision of workers, firms, and landowners. The joint actions of exogenous spatial characteristics and the endogenous economic potential available to agents at each location drive the sorting behavior of firms and the predictions of the model.

Production Environment. A fixed set of entrepreneurs draw a productivity ψ from some distribution $G(\tilde{\psi})$ where $\tilde{\psi} = \psi^{\frac{1}{\sigma-1}}$.[13][14]

Firms are required to locate somewhere in order to produce. Firms employ labor at their locations and pay location-specific wages. The total market of available labor is equal to some mass of L workers. The firms' differentiated goods can be sold to consumers (workers, entrepreneurs, and landowners) at any location. In equilibrium, all firms will choose to service all locations. Access to consumers, local production costs, and rents drive the location decisions of firms.

Landowners use labor to produce units of non-residential space and provide that space to a density of firms.

Productivity spillovers. Each location has a productive amenity $s(i)$:

$$s(i) = \int_S \tau_2(i,j)^{-1} \cdot f_2(t(i), \psi(i), h(i), s(i))dj \tag{1}$$

which may be a combination of an exogenous amenity $t(i)$, an endogenous amenity defined by the quality ($\psi(i)$) and quantity $h(i)$ of firms at i, and an endogenous amenity derived by the firms and amenities close to i. The function mediating the combination of these forces, f_2, is assumed to be bounded and continuous. The decay function τ_2 defines how these forces act through geographic linkages.

Consumption. All three agent types purchase and consume a CES aggregate of all the differentiated goods produced at all locations with an elasticity

[13] Defining the distribution over a function of the productivity parameter rather than the productivity parameter itself simplifies the algebra.

[14] For technical reasons only, the distribution is assumed to have an upper and lower bound, $0 < \psi_L < \psi_U < \infty$, and no other distributional assumptions are made here.

of substitution $\sigma > 1$.[15][16] The quantity of each good consumed will vary in equilibrium by location. Consumers will substitute towards goods produced locally, as such good face lower transportation costs, and goods produced by more productive firms, as the factory prices of such goods will be lower. [17]

Timing. The model is static and all production, location, and consumption decisions are made simultaneously by all agents.

Worker location decision. A mass L of homogenous workers provide one unit of labor inelastically. Workers consume at their location and do not participate in a real estate market.[18] Because all goods will be sold to all

[15]Fixing landowners consumption at their location assumes balanced trade. This assumptions simplifies the notation significantly but does not drive the results. Instead, landowners and entrepreneurs can be assumed to be companies in which each worker owns stock. It is crucial, however, that some consumption remain localized. If all three agents could separate production and consumption choices entirely, then geographic frictions would no longer be a defining element of this model.

[16] Specifically, all three agents maximize their utility

$$U = \left[\int_{\omega \in \Omega} q(\omega)^{\frac{\sigma-1}{\sigma}} d\omega\right]^{\frac{\sigma}{\sigma-1}}$$

where $\sigma > 1$ is the elasticity of substitution and $\omega \in \Omega$ is a good in the set of all available goods Ω.

[17] The local price index at location i is defined as

$$P(i)^{1-\sigma} = \int_{\omega \in \Omega} p(\omega, i)^{1-\sigma} d\omega$$

where $p(\omega, i)$ is the location-specific price of each good.

[18]A more realistic complication models the location decisions and commuting costs of workers. Locations may have differing residential amenities with a third transportation cost function $\tau_3(i, j)$ defining potential commute costs. For such a setup, see Monte et al. (2015). Landowners at each location provide both residential and commercial space, however for the indexing method to work in this scenario, a crucial assumption in this case is that the cost functions of each must be independent of the supply of the other. Because my focus is on the location decisions of firms here, I do not model residential real estate decisions.

locations, workers view locations as varying only by their respective price indexes. For any two locations, homogeneous workers must be indifferent between those two locations. In equilibrium, wages must exactly offset prices such that worker utility equalizes across space.

Firm location and pricing decisions. A set of entrepreneurs receive heterogeneous productivities and will each create a single firm to sell final goods to all agents at all locations.[19] An entrepreneur's location decision affects the variable and fixed costs of production they face as well as the demand they will face. Entrepreneurs face three decisions: (1) whether to produce (2) where to locate and (3) how to price their firm's good.[20] Entrepreneurs simultaneously solve these three problems.

Firm-specific productivity lowers the marginal cost of production which is a function of a location-specific wage:

$$MC(i) = \frac{w(i)}{\hat{\psi}}$$

where observed productivity $\hat{\psi} = \tilde{\psi} \cdot s(i)$.[21]

At any location, the firm's price decision will be a constant markup over its marginal cost,[22] and the pricing decision can be folded into the location

[19] I will use the terms entrepreneurs and firms interchangeably throughout.

[20] The entrepreneur's optimal choice can be found by solving these three decisions in reverse: first determining the optimal price of the good at each potential location, then the optimal location given the pricing rule at each location, and finally whether to produce given the profits yielded by the optimal location.

[21] There are therefore no location productivity advantages arising from access to labor. Introducing commuting costs and residential amenities reintroduces such geographic effects.

[22] For a firm with a given productivity parameter ψ at a given location i, the optimal factory price will be

$$p(\psi, i) = \frac{w(i)}{\rho \cdot \hat{\psi}}$$

decision. Through the wage and productive amenity, marginal costs will vary by location, and the optimal price will follow suit.

Because variable costs and transportation costs to other markets differ across locations, firms face higher or lower demand at different locations. Because consumers substitute towards local firms and locations differ by their productive amenities, demand and variable profits vary across locations.

The firm pays a fixed cost of rent $\phi(i)$ (which is denominated in terms of units of consumption) to the landowner in order to rent space at location i. Rent does not depend on firm size.[23]

Firms face a tradeoff between higher real variable profits and higher real rents. Entrepreneurs' utility is maximized when their real incomes, their firms' profits (in terms of price-index bundles of final goods), are maximized. The firm's maximization function can therefore be written as

$$i^* = argmax_{i \in S}\{\pi_f(i)\} = argmax_{i \in S}\left\{r(\psi, i)/\sigma - \phi(i)\right\}$$

where $r(\psi, i)$ is the revenue of a firm with productivity parameter ψ at location i.

Revenue is a function of the markup, the firm's productivity ψ, the wage at i, as well as the price index at j, $P(j)$, the nominal size of the market at j, $R(j)$, the iceberg transportation costs between i and j, $\tau(i, j)$, and the price index at i, $P(j)$.[24] It is possible to separate the location-specific effects on variable profit from firm-specific productivity:

where $\rho = \dfrac{\sigma}{\sigma - 1}$ is one over the optimal markup.

[23] See Appendix Figure [] for evidence of concavity in commercial rents. While this assumption could be disposed with, further assumptions on the supermodularity between firm quality and location quality would then have to be made.

[24] The revenue, in units of consumption at i, from selling from point i to point j:

$$r_j(\psi, i) = \frac{\left[\dfrac{w(i) \cdot \tau_1(i, j)}{P(j)\rho\hat{\psi}}\right]^{1-\sigma} R(j)}{P(i)} .$$

$$r(\psi, i)/\sigma = \psi \cdot \eta(i) \tag{2}$$

where $\eta(i)$ is the location-specific advantage term defined as

$$\eta(i) = \int_{j \in S} \frac{w(i)^{1-\sigma} \tau_1(i,j)^{1-\sigma} R(j)}{\sigma \rho^{1-\sigma} P(i) P(j)^{1-\sigma} \cdot s(i)^{1-\sigma}} dj.$$

The value of $\eta(i)$ is dependent on the variable cost of production at i and the transportation cost-weighted proximity to markets j, both a function of the market size of j and the price index at j. Several centripetal and centrifugal geographic forces are embedded. Proximity to high-productivity firms increases affects advantage by (1) increasing the size of local markets, (2) stiffening competition, through lower price indexes in nearby locations, (3) decreasing marginal cost of production through changes in $w(i)$ or $s(i)$, and (4) increasing entrepreneur utility. The first two channels operate through market access,[25] while the last two operate through effects on the

Summing over all markets $j \in S$, the expression for the variable profit at point i :

$$r(\psi, i)/\sigma = \int_{j \in S} \frac{w(i)^{1-\sigma} \cdot \tau_1(i,j)^{1-\sigma}}{P(i) P(j)^{1-\sigma} \rho^{1-\sigma} \psi \cdot s(i)^{1-\sigma}} \frac{R(j)}{\sigma} dj$$

The above equation yields the relationship between a firm's variable profit and its location.

[25]The market potential at i can be expressed as:

$$Market\ Potential\ = \frac{\eta(i) P(i)}{w(i)^{1-\sigma} \cdot s(i)^{1-\sigma}} = \int_{j \in S} \left[\frac{\tau_1(i,j)}{\rho P(j) \cdot s(i)} \right]^{1-\sigma} \frac{R(j)}{\sigma} dj.$$

The parameters of the utility function, the equilibrium distribution of firms and workers, and the exogenous geography (through τ_1) jointly determine a location's market access. Locations that are relatively proximate to more productive firms and far from less productive firms will have lower price indexes, which decrease the effective market for a given firm. On the other hand, larger firms, their workers and landowners together form larger markets, and relative proximity to these markets increases market access via $R(j)$. Together with the variable costs at a given location i, these forces govern location i's access to markets.

price index and the endogenous productive amenity at location i.[26]

In equation (2), the ability to separate location- and firm-specific contribution to variable profit is a result of the particular assumptions of the production function. In particular, two features of equation (2) are crucial.

First, higher η locations have higher variable profits for firms holding ψ constant. All firms agree on which locations yield the highest variable profit. This ranking orders every location $i \in S$ according to $\eta(i)$. For any two locations i and j, for any given level of firm productivity, marginal profits will be higher where η is higher. The first proposition restates this:

> *Lemma* 1 : For any two locations η_1, $\eta_2 \in [\bar{\eta}, \underline{\eta}]$, if $\eta_1 > \eta_2$, then $r(\psi, \eta_1) > r(\psi, \eta_2)$ i.e., variable profits at η_1 are higher for all firms.

Lemma 1 follows directly from the production function and definition of η.

Second, $\hat{\psi}$ enters multiplicatively with η. As a consequence, difference in variable profit between any two locations is higher for more productive firms.

> *Lemma* 2 : For any η_1, $\eta_2 \in [\bar{\eta}, \underline{\eta}]$, if $\eta_1 > \eta_2$, and $\psi_1 > \psi_2$, then $r(\psi_1, \eta_1) - r(\psi_1, \eta_1) > r(\psi_2, \eta_2) - r(\psi_2, \eta_2)$.

Lemma 2 also follows directly from equation (2) and is a result of the supermodularity between location and productivity assumed in the production function. Locations that have higher equilibrium market access will be more sought after by more productive firms. As in the assignment literature,[27] this condition will, in equilibrium, lead firms to sort on productivity, with more productive firms capturing locations with a combination of greater market access and lower marginal costs.

[26] Proximity to productive firms reduces the price index at i, which reduces the wage $w(i)$, and thus marginal cost at i, and directly increases entrepreneur utility by increasing the total amount of consumption for a given level of profits. Proximity also directly effects productivity at i through $s(i)$.

[27] Costinot & Vogel (2009) use log-supermodularity to ensure matching.

The equilibrium parameter η and the ability to rank any $i \in S$ according to η will be central to the subsequent analysis. It is important at this stage to reiterate that the mapping of η into locations is defined endogenously; it will be dependent on a particular arrangement of firms and workers in the space. As the analysis above demonstrates, the equilibrium decision of each firm depends on the decisions of all other firms. Prices, market access, and spillovers at any given location i will be impossible to pin down without knowing those variables at every other location.

Finally, firms produce if their profits, conditional on their optimal location decision, are above zero:

$$r(\psi, \eta^*)/\sigma \geq \phi^*$$

where η^* is the η of the profit maximizing location for firm with productivity ψ, and ϕ^* is the rent at that optimal location.

Landowner supply decision. Each landowner is endowed with a location and must decide how much density to provide to firms at that location. Density is provided at increasing marginal construction cost according to an invertible, twice-differentiable cost function $c(h)$:

$$c'(h),\, c''(h) > 0,\ c(0) = 0$$

where h is the density of firms at a particular location. Construction costs are denominated in baskets of final goods and no labor is required in construction.[28]

[28]Real construction costs differ from location to location due to differences in the price index. Landowners at higher $P(i)$ locations face higher unit costs of building, as real wages are higher at such locations. However, they also receive higher real rents (controlling for η since $\phi(\eta)$ is denominated in units of final goods and therefore real rents are higher, all else equal, at locations with higher $P(i)$ as well. Therefore I write the cost function $c(h)$ as denominated in units of final goods. Thus, the landowners' provision decision is homogenous degree zero with respect to $P(i)$, and can be made entirely in terms of real units of goods. Denominating construction costs in final goods simplifies the algebra of the labor market clearing condition. Alternative specifications with costs denominated in labor are possible.

Landowners tradeoff the price-adjusted costs of providing density $c(h)$ against the price-adjusted rents ϕ. In equilibrium, the landowner chooses rent ϕ and density h to maximize profits

$$\pi_l = h\phi - c(h).$$

The landowner's choice of h is unconstrained. However, the choice of ϕ is constrained by the participation constraints of the firms.[29] The landowner will choose the highest possible rent conditional on some firm type being willing to locate at her location. This is equivalent to choosing the firm with the maximum willingness to pay for her location.

$$\pi_l = max_{\phi} \{h\phi - c(h)\}$$

$$\text{s.t. } \phi \leq \max_{\psi} \{\min_{\eta' \in S \not\eta} \{\psi \cdot (\eta - \eta') + \phi'\}\}.$$

Notice that landowners differ only according to their location parameter η. In the equilibrium, the decisions of landowners and firms will jointly determine the matching function $\psi(\eta)$ between firm-types and the location index.

[29]Thus, the landowner's constrained optimization chooses highest rents possible, conditional on the willingness to pay of firms. Each firm's willingness to pay for a space at location with value η will be conditional on the firm's outside options, including the rent ϕ' at outside options $\eta' \in S \not\eta$, and variable profits at η and locations η'. A firm of productivity ψ's willingness to pay for space at location η can be expressed as

$$WTP(\psi, \eta) = \min_{\eta' \in S \not\eta} \{\psi \cdot (\eta - \eta') + \phi'\}.$$

Intuitively, given a single outside option η', a firm is willing to pay rent at η equivalent to the difference in variable profit that firm would collect at location η and what would be the variable profit at location η', paying whatever rent is charged by the landowner at η'. The firm's willingness to pay for η is therefore the minimum difference between profits at η and profits at all other outside options $\eta' \in S \not\eta$, taking rents at those locations as given.

3. Equilibrium

I introduce η as an index variable and define the rent function $\phi(\eta)$, establishment density function $h(\eta)$, employment density function $l(\eta)$, and the matching function $\psi(\eta)$. An equilibrium is then characterized by six equilibrium conditions: (1) the firm spatial equilibrium, (2) the land development equilibrium at each location, (3) the real estate market clearing condition, (4) the labor market clearing condition, (5) the worker spatial equilibrium, and (6) the goods market clearing and balanced trade condition.

The analysis herein will first assume an assignment $\eta(i)$, uses the first three conditions to solve for $\phi(\eta)$, $h(\eta)$, $l(\eta)$, and $\psi(\eta)$, and the last three to solve for $\eta(i)$. In order to demonstrate existence, it must be shown further that there exists a mapping of index values to actual locations. This final mapping pins down the general equilibrium: prices, market access, and the size of productivity spillovers in every location. The mapping is a system of nonlinear integral equations resembling the equilibrium conditions in Allen & Arkolakis (2013).[30] Crucially, this final step provides no analytic solution and therefore yields no additional predictions.

Index variable. I begin by positing a mapping $\eta(i)$ which characterizes all occupied locations according to η. η serves as index that transforms the space of all occupied locations[31] into a space $[\underline{\eta}, \bar{\eta}]$, with boundaries defined by the maximum and minimum values of η in a given equilibrium. For each value of $\eta \in [\underline{\eta}, \bar{\eta}]$, a certain density of locations, expressed as $f(\eta)$, will share this value. [32]

[30] I show that under the particular assumptions of this model, an equilibrium must always exist. Uniqueness is guaranteed under particular conditions for transportation costs that are given in Appendix 3. Because these conditions may not be met, and because the imposition of uniqueness generates no further predictions in this context, I do not impose them.

[31] Given the assumptions of the model, in particular that $c(h)$ is continuous and $c(0) = 0$, the equilibrium must always be regular and this transformation is always $S \rightarrow [\underline{\eta}, \bar{\eta}]$.

[32] As an illustration, consider a two-dimensional circular geography with a unit radius, where transportation costs are linear with distance traveled, and all goods must travel through the center of the circle. In this geography, locations closer to the center always have lower transport costs to all other locations and therefore in any equilibrium, η will be decreasing with distance to the center, with all locations on a fixed radius away from the center sharing the same index value η. For any radius r,

Rent gradient and firm sorting. Next, I introduce the rent function $\phi(\eta)$, characterizing the rents at each location according to the location index. Because the landowners' optimization problems differ only by η, landowners with identical η's choose identical rents; η is a sufficient statistic for ϕ.

In a spatial equilibrium, no entrepreneur can increase her implicit utility by changing locations. Deviations from optimal locations could potentially increase profits in one of two ways: an entrepreneur could move to a location with higher variable profits, or to a location with lower rent. Restating the firm's profit maximization implies neither of these options increase real profits for any entrepreneur.[33] Formally,

$$\forall \psi \in \Psi,\ \phi(\eta) - \phi(\eta_L) \leq \psi\left(\eta_L - \eta\right). \forall \eta_L < \eta \tag{3}$$

$$\forall \psi \in \Psi,\ \phi(\eta) - \phi(\eta_H) \geq \psi\left(\eta_H - \eta\right),\ \forall \eta_U > \eta \tag{4}$$

where η is the location chosen by entrepreneur of productivity ψ, η_L is an outside option with $\eta_L \leq \eta^*$, and η_H is an outside option with $\eta_H \geq \eta^*$. Equations (3) and (4) are the incentive compatibility constraints for firms optimally locating in spaces with an index value of η.

It immediately follows from equations (3) and (4) that $\phi(\eta)$ is strictly

$$\eta(r) = (1-r)\bar{\eta} - r\underline{\eta}$$

with a density function $f(\eta)$ defined as

$$f(\eta) = 2\pi\frac{\bar{\eta} - \eta}{\bar{\eta} + \underline{\eta}}.$$

[33] The firms' basic tradeoff between variable profit and rent is in terms of baskets of goods so that real profits and implicit utility are equivalent.

increasing.[34]

Next, I rewrite the entrepreneur's optimization problem in terms of η and the profit function $\phi(\eta)$,

$$\pi_f = max_{\eta \in [\underline{\eta}, \bar{\eta}]} \{\psi \cdot \eta - \phi(\eta)\} ,$$

which yields the first order condition

$$\psi(\eta) = \phi'(\eta) \; \forall \eta \in [\underline{\eta}, \bar{\eta}]. \tag{5}$$

Equation (5) defines the relationship between the mapping of firms ψ to locations η to the rent gradient.[35] The interdependence between these two illustrates a deeper intuition of the model, that, over and above differences in location quality, the rent gradient reflects the underlying variation in firm quality. This is a departure from the standard results of the AMM and Rosen-Roback frameworks, where rents solely reject differences in location quality. Rather, in this context, understanding the rent gradient requires understanding of the sorting pattern of firms.[36]

[34] If, instead, a local minimum existed at $\eta_1 \in (\underline{\eta}, \bar{\eta})$, all firms at locations to the left of η_1 could capture higher variable profit at locations with reduced rent. The firms' incentive compatibility constraints would then guarantee such locations would be unoccupied, contradicting the definition of all locations in $[\underline{\eta}, \bar{\eta}]$ as the set of occupied locations.

[35] This condition can also be derived from the incentive computability constraints using the Mirrlees conditions. Following Mirrlees (1976), we can further restrict the relevant outside options $\eta_H, \; \eta_L$ to be locations with the value of η closest to η^*. Intuitively, if the firms have optimized, deviations from the optimum are increasingly detrimental, and therefore attention can be restricted to local deviations. Thus, for any choice set over which η is continuous, equations (3) and (4) can be evaluated by taking the limit as $\eta_H \to \eta_L$. In the limit, both inequalities bind, becoming the mapping function in equation (4). In this way, the matching between firms and landowners is analogous to a mechanism design problem with a continuum of types with private information.

[36] Appendix 2 verifies that given a matching function $\psi(\eta)$ and rent gradient $\phi'(\eta)$, landowners at every $\eta \in [\underline{\eta}, \bar{\eta}]$ choose rents such that equation (4) is satisfied. Because (4) is derived from the firm's optimization, and firm's optimization is the constraint on the landowner's price decision, it exactly satisfies the latter.

Finally, note that for firms to be at their optimum, the second order condition $\phi''(\eta) > 0$ must hold, which together with equation (5) implies that $\psi'(\eta) > 0$. This last result guarantees the positive assortative matching between firms and locations, and as expressed in the previous sub-section, is a product of the super-modularity of the production function. Proposition 1 formalizes this result

> $Proposition\ 1$: **Firm sorting.** In any equilibrium, a strictly increasing function $\psi(\eta)$ exists; its inverse $\eta(\psi)$ is weakly positive. That is, in any equilibrium of rents, firm, and worker locations, for any $\eta_1 > \eta_2$, $\psi(\eta_1) > \psi(\eta_2)$. Furthermore, in any equilibrium of the above model, a non-degenerate distribution of $\eta(i)$ must exist for some locations in S. The only stable equilibria exhibit "strong" sorting, such that the one-to-one function $\psi(\eta)$ is continuously increasing.

Appendix 1 first proves the assumptions of the model prohibit a degenerate equilibrium distribution of $\eta(i)$, then that a location with a higher η must be captured by firms with higher productivity parameter ψ. [37]

Density gradient. Next, I rewrite the firm's profit maximization condition using the rent gradient

[37] The appendix further elaborates on the possible scope of exceptions to sorting. Any equilibrium must display "weak" sorting properties: the function $\psi(\eta)$ must be weakly increasing. When the function $f(\eta)$ is non-singular, sorting is strict. Singularities in $f(\eta)$ imply a positive mass of locations with a single value of η, and therefore a range of firms at that value of η. However, two firms of the same productivity may not be found at locations with different η's. The appendix also shows that any weak sorting equilibrium is unstable.

The remaining analysis assumes a stable one-to-one matching of firms to location productivity. The appendix also notes small changes to the leading predictions of the model in the case of a weak sorting equilibrium, and shows the corollary to Proposition 1, that uniform density is never an equilibrium characteristic, and therefore agglomeration is a pervasive characteristic of this model.

$$\tilde{\pi}_l = \max_h \{h \cdot \phi(\eta) - c(h)\}.$$

The first order condition of the above equation implicitly defined the density provision of landowners as a function of η :

$$h(\eta) = c'^{-1}(\phi(\eta)). \tag{6}$$

Equation (6) sets the density of firms at a given location i, given the type of firm at location i, such that the cost of accommodating the marginal firm is equal to the willingness of that firm to pay for space at i. While the firm's optimization pins down the rent gradient, the landowner's optimization adjusts the density of firms at each point, thereby distributing the mass of firms in a space according to the index η. The density function $h(\eta)$ moves the mass of firms from the space ψ, where the density is defined according to the distribution $g(\psi)$ to the index's space.

Because of the assumptions on the cost function, $h'(\eta) > 0$, and establishment density is increasing in η. In turn, landowners in more productive locations reap larger real profits, both because they attract more productive firms with higher willingness to pay, and because they optimally accommodate a higher density of firms.

Real estate market clearing condition. The previous two equilibrium conditions are derived conditional on a mapping of $\hat{\psi}$ into η, which moves the distribution of firms by productivity into a distribution of firms on the space of η. The landowners' optimal provision of density will feed, in equilibrium, back into the assignment of firms to locations, as more density at some locations shifts the mass of firms towards those locations. The matching of firms to locations and rents adjust accordingly so as to accommodate all firms which choose to produce.

Each ψ has associated with it a specific density of firms, $g(\psi)$. The density function $h(\eta)$ describes the equilibrium density of firms at each location according to η. The function h and g are defined on different spaces and their distributions will not necessarily resemble one another.

However, the total mass of firms choosing to produce must equal the total mass of firms accommodated equilibrium, $H(\eta)$. Furthermore, this holds for any ψ,[38] and therefore

$$h(\eta) = \frac{g(\psi(\eta))\psi'(\eta)}{f(\eta)} \tag{7}$$

Equation (7) relates the mapping $\psi(\eta)$ to the firm density function $h(\eta)$ and the density function $f(\eta)$. For any given allocation of firms to locations, the initial density of firms of a specific productivity type must exist somewhere in the actual space.

Employment Density. I further define employment density as a function of location advantage: $d(\eta)$. The equilibrium size of a firm and is described by the equation

$$l(\eta) = \tfrac{\sigma \cdot \rho}{\lambda} \cdot \psi(\eta) \cdot \eta.$$

Using the above equation, labor market clearing condition expressed with respect to η and ψ, the equality

$$\int_{\underline{\eta}}^{\eta} d(\eta) \cdot f(\eta) d\eta = \int_{\underline{\psi}}^{\psi} \left[\frac{\sigma \cdot \rho}{\lambda} \cdot \psi \cdot \eta(\psi) \cdot g(\psi(\eta)) \right] d\psi$$

must hold for each variable upper boundary $\psi(\eta)$ and η. Differentiating both sides and substituting equation (6) yields an equation relating employment density to the previously defined functions:

$$d(\eta) = \frac{\sigma \cdot \rho}{\lambda} \cdot \psi(\eta) \cdot \eta(\psi) \cdot h(\eta).$$

[38] The total mass of firms with at least a given productivity ψ must be the mass of firms with at least ψ in equilibrium. This intuition yields the following equilibrium constraint

$$\int_{\underline{\eta}}^{\eta_j} h(\eta) f(\eta) d\eta = \int_{\psi(\underline{\eta})}^{\psi(\eta_j)} g(\psi) d\psi$$

at any location j. Since this condition holds everywhere, differentiating at $\eta(j)$ we find equation (7).

General Equilibrium. While equations (5)-(7) describe an equilibrium in the space of η, they do not solve for the general equilibrium, which must pin down prices, the distribution of labor, and spillovers at each location. To do this, the model must pin down a mapping of values of η to all locations $i \in S$.

$Proposition\ 2$: An equilibrium exists and is described by equations (A3.8)-(A3.11).

Appendix 3 uses the labor market clearing condition, the balanced trade condition, and equations (5) - (7) to express that mapping and proves the system exhibits at least one nontrivial solution and provides conditions for uniqueness. However, because the uniqueness conditions are not easily verifiable, and because they don't on their help provide predictions, I will not assume they are met through the remainder of the paper.

4. Derivation of Observable Sufficient Statistics

In this section, I give conditions under which the location index η can be proxied for using observables. The advantage of the strategy of building a index rests in the ability to do so. As in other geography models,[39] the interdependence of geographic decisions among many agents–in this case firms, workers, and landowners at each location–does not yield a closed-form solution and creates the potential for multiple equilibria. Because of these features, no predictions can be made for the relationship between geographic variables and location or firm characteristics.

However, when one-to-one mappings between both the index and location observables, locations can be ordered according to these observables and therefore in a manner which captures geographic linkages. At the same time, the new ordering abstracts from the physical structure of economic activity and geographic linkages, so that this ordering, for example, can equally interpret monocentric and multi-centric urban spaces.

Using this and the equilibrium conditions derived in section 3, we can

[39] See, for example, Allen & Arkolakis (2013).

formally state the relationship between these three observables and the unobserved η:

> $Lemma\ 3$: The rent gradient is increasing and convex $\partial^2\phi(\eta)/\partial\eta^2 > 0$. Firm density is increasing in η, $\partial h(\eta)/\partial\eta > 0$. Employment density is increasing in η, $\partial d(\eta)/\partial\eta > 0$

Lemma 3 follows directly from the first and second order conditions of the firm and landowner optimization problems. As ensured by super-modularity, more advantageous locations are matched to more productive firms. From equation (4), it is clear that this alone ensures the rent gradient is increasing and convex in η. Because, by equation (4), the rents are increasing in η, and landowners provide density such that the increasing marginal cost equals the rent, more advantageous locations must have higher density. Proposition 3 follows directly from Lemma 3:

> $Proposition\ 3$: The relative values of any of the three observables firm density, employment density, and rents, between any two locations reveal the relative values of η at those locations.

Three assumptions have been crucial in deriving this result for firm and employment density. First, the construction cost function $c(i)$ is constant across locations. If this were not the case, rents, but not densities, would continue to reveal the relative rankings of η.[40] Second, rents are assumed to be paid as a fixed and not as part of a variable cost. Were this not the case, firm density would no longer function as a sufficient statistic for η.[41] Third, firms do not compete with workers for real estate. Were this not the case, neither density would function as a sufficient statistic for η.[42]

[40] The first order condition of the density provision decision shows that areas with higher marginal cost would have lower employment and firm density. However rents, subject to competition among locations, would be continue to be increasing in η. This is no longer true if landowners are large enough to be quasi-monopolistic.

[41] See footnote 26 for a discussion of this scenario

[42] See footnote 17 for a discussion of this scenario.

To the extent that these assumptions do not hold in the data, the density observables will fail to function as sufficient statistics for η and so will not capture the full effect of geographic linkages. However, evidence in the next section will show that despite these limitations, observable densities function better than rents and other measures of location quality.

Finally, the use of density as a sufficient statistic provides a check for the stability of the equilibrium across time periods. The model yields multiple equilibria, however the equilibrium is conditionally unique with respect to the index η. In particular, if, when comparing two equilibria between any two values of $\eta \in [\eta_1, \eta_2]$, $f(\eta)$ is constant and $\psi(\eta_1)$ is constant, then, as is apparent from Equation 8 in Appendix 2.2, the mapping function $\psi(\eta)$ is constant across both equilibria. All comparative statics are constant across both equilibria between η_1 and η_2. Verifying that $f(\eta)$ between η_1 and η_2 and $\psi(\eta_1)$ is constant between time periods, or equivalently that $f(h(\eta))$ between h_1 and h_2 and $\psi(h(\eta_1))$ is constant between time periods is sufficient to guarantee that any observed change is a test of a comparative static and not a jump across equilibria.

5. Application I: An empirical alternative to AMM

This section shows how the observable sufficient statistics can be used to reorganize urban geographies in a way which better accounts for the geographic linkages between locations. I derive predictions for the quality of firms at each location based on the relationships in the model between firm qualities and the sufficient statistics for location advantage. Rather than define points in a city according to their distance to a central point (in the AMM model, a central business district), locations can be defined by their observable stand-ins for location advantage.

To test wether geographic linkages matter, I compare the resulting empirical relationships to results from a baseline AMM model, where firms sort according to their distance to the city center. For Business Services firms, comparable elasticities are recovered. However, for manufactures, elasticities from the sufficient statistics cannot be replicated using the AMM model.

I also combine both measure in a single regression order to confirm that employment and establishment density can be used as sufficient statistics and find that density metrics, but not distance measures, may be used as sufficient statistics.

Predicted relationships between firm and location characteristics. In the set of equilibria derived in Section 3, there is no closed-form solution to the mapping of locations i to the index η. Despite this, because the location index is a sufficient statistic for both the locations' characteristics and the characteristics of firms locating there, the indexing method yields predictions. Equations (4)-(10), generate relationships between observable location and firm characteristics and the location index. I show how predictions on key relationships between firm and location characteristics can be derived from these equations. Appendix Table 2 summarizes these relationships and Appendix 4 derives them.

This approach only bears predictions for variables that can be expressed as monotonic functions of η. Prices, market access, and productive amenities constitute the geographic components that form η, and can only be pinned down in a general equilibrium.

Because more productive firms sort into higher η locations, firms in denser locations will have higher ex-ante productivity. However, this alone does not guarantee an observed relationship between density and productivity. If firms prefer to forego spillovers at certain locations for market access at others, higher η locations will be those with lower spillovers and higher levels of market access. If differences in $s(i)$ are greater than underlying differences in $\psi(\eta)$, observed productivity will be higher at worse locations. In a model where firms choose among several dimensions, supermodularity between spillovers and productivity is not enough to guarantee the density-productivity relationship.[43]

[43] Appendix A4.1 gives conditions under which observed productivity will also be increasing in location density. Intuitively, if either market access or spillovers are weak enough, firms sort on one but not the other, and the observed relationship emerges. Only when spillovers are weak enough to give up but strong enough to reverse underlying differences in firm productivity will the relationship reverse.

The model predicts firms in denser locations will be larger, both because these firms were ex-ante more productive, and because firms at those locations face higher demand through combination of higher productivity and higher market access.

Finally, the model predicts that firms in denser locations will be more profitable. Higher profits are due to higher ex-ante productivity and the because landowners cannot capture the full impact of location quality on profits.[44] This prediction can only hold in models of firm sorting.[45]

Empirical results and comparison to AMM model. I estimate the following equation:

$$log(F_{it}) = \alpha_0 + \alpha_1 \cdot log(T_{tr,t}) + \alpha_2 \cdot X_{c,t} + \alpha_3 \cdot X_{i,t} + \epsilon_{it}$$

where F_{it} is the establishment-level characteristic for establishment i measured at time t, $T_{tr,t}$ is the tract level characteristic for tract tr , measured at time t, $X_{c,t}$ is a vector of city by industry by year fixed effects,[46] and $X_{i,t}$ is a set of establishment and tract variables at time t, including age, and cubic polynomials of the establishment's latitude and longitude to account for potential spatial auto-correlation in the data.

Tables 1 reports relationships based on the first two predictions, productivity and employment, for establishments in all industries. [47] Column

[44]See footnote 24 for an explanation based on results from the mechanism design literature.

[45] Specifically, in models with agglomeration forces and homogenous firms, spatial equilibrium holds that price differences, wage differences, and location productivity must offset each other so that all firms are indifferent between locations. Firms pay for increased productivity with higher location-specific costs. If costs did not offset productive amenities, and profits differed across locations, firms in other locations could do better by relocating. In such models, differences in profitability can only be maintained if entrepreneurs receive location-specific consumption amenities that are higher in unprofitable locations so that entrepreneurial utility equalizes across locations when profits do not.

[46]The model is constructed around a single industry, so industry fixed effects remove industry-specific variation. Because my sample spans the 2002 SIC / NAICS crossover, I use NAICS-year or SIC-year fixed effects for each code. However, results do not change in sign or significance when industry fixed effects are omitted. For TFP results, the relevant elasticities increase tenfold when industry is omitted.

[47]I use the employment density measure which is calculated by measuring employment at the tract level, minus employment at the establishment, divided by the square miles of

one and four shows that firms in denser locations are more productive and larger, measured by log output per worker and log employment at the establishment level, respectively. These elasticities are consistent with the literature.[48] Columns two and five repeat the exercise using tract distance to CBD, and find similar elasticities.[49]

Columns three and six combine the distance and density metrics. The addition of distance metric does not affect the coefficient on density, while the converse is not true. When comparing the marginal r-squared of the two tract-level characteristics, substantially more variation is explained by the density metric. These results are consistent with the hypothesis that density is a sufficient statistic for the firm characteristics at a location and inconsistent with the hypothesis that distance to CBD can be. Nevertheless, it appears that for these measures, the additional information gained using the density measures does not alter results.

A binned scattered residual in figures 1 and 2 further illustrate. Each bin is a percentile of tract proximity or density, residualized for the controls in Table 1. Most of the positive variation in the density gradient relationships are derived from tracts extremely close to the geographic center of the CBD or in extremely distance areas. By contrast, the density relationship is consistent throughout.

However, when results are broken down by industry, this is no longer the case. Table 2 reports these and other measures available for manufactures establishments. The distance metric fails to predict relationships between location and three measures of productivity,[50] as well as gross margin per

the tract. Nearly identical results can be found using establishment density. All regressions control for establishment-level controls, including age of establishment and a dummy for establishments existent as of the 1976 LBD, before which no longitudinal data exists and cubic polynomials for latitude and longitude, as well as 6-digit NAICS or 6-digis SIC by year by MSA fixed effects. All subsequent analysis shares these controls in order to directly compare firms within the same industry and the same city and year. The model does not require MSA boundaries and indeed losing the MSA fixed effects does not alter coefficients for the density elasticities.

[48] See Rosenthal & Strange (2004b) for a complete review of previous findings.

[49] Negative elasticities denote tracts further away from the CBD are less productive and smaller.

[50] Log output per worker, log value added per worker and Haltiwainger TFP measures. The

worker, a measure of marginal cost and profitability. Distance has the opposite prediction on employment, and gross margin per worker, a measure of profitability and marginal costs, are not explained by distance but continue to be correlated with the density metric, in line with the model.

Taken together, the results of Table 2 show that the distance measure fails to describe the geographic description of manufactures, which appear not to organize with respect to central business districts, even while larger, more productive manufactures continue to locate in denser areas. While the distance metric is marginally outperformed when including business services,[51] the divergence between the distance and density metric implies that the monocentric model fails to capture the geographic patterns of manufactures.

Appendix Table 3 further explores the information content of these competing metrics by imposing a factor model on the data. To the extent that a single underlying variable can explain variation in establishment characteristics, measures of density appear to be much more strongly correlated with this underlying variable.

6. Application II: Tests for the existence of firm sorting

Under the sorting hypothesis, the productivity of firms at a given location is a function of the mapping $\psi(\eta)$, itself a function of the rent and density at all other locations Exogenous changes to the provision of floor space at a single location can affect the entire distribution. In the following subsection, I examine a specific instance where small, positive shocks to density generate changes to local productivity through composition effects. Specifically, a shock to floor space in any tract generates negative effects on the produc-

elasticity for the density departs from the literature on city size due to the detailed industry controls. Note that the majority of the variation is explained using industry controls, and while the output per worker elasticity does not decrease when controlling for industry, the elasticity on TFP does. This suggests that the majority of the geographic distribution of TFP and the relationship between TFP and density is mostly due to the industrial composition across space.

[51] Appendix Table 2 reports these results for the sample including just business services. Results are robust to this subsample.

tivity of entering firms in less advantageous locations and positive effects in more advantageous locations.

Intuitively, the new firms at a positively shocked location may come from either better or worse location, but will be marginal in either. All firms that would have entered at worse locations are priced out of the more advantageous locations. Those now entering those locations from below will be those who on the margin, would have been the most productive firms at the lesser location. The reverse is true for firms entering from above: the firms most easily enticed into the new space available at the less advantageous location must have been the most marginal, and therefore least productive in the better location. The removal of marginal firms from the upper and lower margins negatively and positively affect the quality of the average firms in worse and better locations respectively.[52]

> $Prediction\ 6$: If productivity spillovers are sufficiently weak, increases in construction between any locations $[\eta_{c1}, \eta_{c2}]$ reduces average productivity $\tilde{\psi}_{NL}$ in the set of less-advantageous neighboring locations $[\eta_{c0}, \eta_{c1}]$, $d\tilde{\psi}_{NL}/da \leq 0$. The reverse is true for more advantageous locations, $d\tilde{\psi}_{NU}/da \geq 0$.

When productivity spillovers are strong, the prediction becomes ambiguous. If spillovers are based on nearby density, the productivity of nearby firms could increase, and the sorting and spillover channels would be at odds. If spillovers are based on local firm productivity rather than density, spillover and sorting forces act as complements and the size of the change in productivity will be amplified by decreases in the spillover. However, a negative relationship between available space and productivity cannot be obtained without sorting.

In a model where proximity to density increases productivity, the additional density should positively affect productivity both at higher and lower

[52]Because only one or both margins of change may be in effect for the productivity boundaries of $[\eta_{c1}, \eta_{c2}]$, the signs of the predictions in Prediction 6 is weak.

density neighboring locations, holding the density at those locations constant. The differential impact on higher and lower indexed locations is unique to this model.

To test Prediction 6, I use the ordering of tracts by density to estimate the following tract-level regressions

$$log(\tilde{\psi}_{NL,t}) = \alpha_0 + \alpha_1 \cdot log(cont_{tract,,t-1}) + \alpha_2 X_{tract} + \alpha_3 \cdot X_t + \epsilon_{tract,t}$$

$$log(\tilde{\psi}_{NU,t}) = \alpha_0 + \alpha_1 \cdot log(cont_{tract,,t-1}) + \alpha_2 X_{tract} + \alpha_3 \cdot X_t + \epsilon_{tract,t}$$

where $\tilde{\psi}_{NL,t}$ and $\tilde{\psi}_{NU,t}$ are the average productivity of entrants to the next-lowest and next-highest ranked tracts, respectively, X_{tract} is a vector of tract-level variables including (and limited to at first) tract fixed effects and X_t are year fixed-effects. Because firm density is a sufficient statistic for location advantage, I rank each tract according to the establishments per square mile in each in 1992, the first year in my data.

Columns one and two of Table 3 report the OLS results of construction at more advantageous locations. Increases in construction expenditures have negative effects on the productivity of entrants at lower-ranked tracts, consistent with the sorting hypothesis. While this coefficient is small, because doubling construction only increases local density by about a quarter, and because nearby firms are similar in productivity, this coefficient is large relative to the elasticity between productivity and density. However, because sorting may interact with agglomeration forces, this elasticity cannot be used to interpret the relative strength of the forces without additional structure on the nature spillovers. Controlling for changes in establishment density, in column 2, does not meaningfully change the coefficient.

Column three and four replicate the exercise for construction expenditures at lower-ranked tracts. The OLS results are similar in magnitude but positive, consistent with the sorting hypothesis. Also consistent with Prediction 6, there is a marked differential effect on entrants when construction expenditures increase at higher vs lower ranked tracks. Shocks to the amount of space provided in a particular location differentially affect other

tracts based on their relative advantage. Taken together, results from these columns confirm the suggestive evidence of intra-city firm sorting.

Columns five through eight repeat the exercise using distance to CBD as an organizing principle for the city. The estimate are insignificant and only sometimes the correct sign. Now construction expenditures appear to have no effect on the quality of entrants and there is no evidence of firm sorting. This strongly suggests that the results of the first four columns are sensitive to the organization of tracts within the city, and that the organization suggested by the AMM model loses critical information.

Table 4 repeats the test in Table 3 but includes construction at all other tracts in a CBSA. Column one reports the results of construction expenditures at all denser tracts in the CBSA. Construction expenditures in denser tracts decreases the productivity of entrants on average. In column two, construction expenditures at less-dense tracts increases the average productivity of entrants. Both of these signs are consistent with Prediction 6 and sorting. Columns three and four report that, just as in Table 3, the AMM rankings do not give evidence of sorting.

Using the instrument for construction expenditures discussed in Appendix 7, the last two columns of Table 4 repeat the first composition effect test (for density percentiles) using predicted construction expenditures as an instrument. The coefficients increase in size but remain differentially negative and positive.

7. Conclusion

The location index approach advanced in this paper bridges two distinct approaches to spatial frictions. Economic geography, which attempts to account for the effects of spatial frictions in a world where locations are networked by their geographic interconnections, and urban economics, which abstracts from most forms of geographic interconnections. Because spatial equilibria in urban economics are divorced from complex notions of proximity, equilibrium conditions deliver clear, testable predictions. In economic geography, technical issues associated with geographic linkages inhibit mod-

els from delivering such clear predictions. This new approach allows for geographic linkages to matters but circumvents these technical issues and delivers reduced-form predictions.

Using observable sufficient statistics, the new theoretical approach also creates a new empirical strategy and organizing principle for reduced-form geographic work, especially intra-urban empirical work. By sorting locations according to location density, locations within cities no longer need be organized according to their distance to a central business district. This new organization allows us to assess other shorthands that have been developed for proximity. In particular I find that the monocentric model performs sufficiently well as an organizing principal for business services firms but not for manufactures. It also reveals the first reduced-form evidence in favor of the firm sorting hypothesis.

The existent evidence against firm sorting at the inter-city level should be reevaluated within the context of within-city sorting. Combes et al (2012), for example, do not examine whether intra-city heterogeneity can explain the apparent lack of pattern in the productivity cutoffs of firms or the thicker right tail of firm productivity among larger cities. It may well be that sorting within cities replicates the patterns observed in the data.

Finally, although this paper did not seek to estimate the strength of market access, its intuitions reconcile two literatures that have previously done so. Papers in the geography literature such as Ahlfelt et al. (2012) have estimated large impacts of changes in market access on the distribution of economic activity. Papers in urban economics that attempt to measure the relative strength of market access such as Ellison & Glaeser (1994)have been more skeptical of the continued importance of location to markets. The results here offer a possible solution: while real differences in market access may be small, they may interact with real estate market elasticities and the sorting behavior of heterogenous firms to create large differences in the distribution of economic activity.

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Figures and Tables

Table 1. Establishment characteristic gradients, all industries

	Log sales per worker			Log employment		
	(1)	(2)	(3)	(4)	(5)	(6)
Log employment density, tract	0.0515**		0.0503**	0.0799**		0.0825**
	(0.0025)		(0.0029)	(0.0023)		(0.0027)
Log miles to CBD, tract		-0.0460**	-0.0072		-0.0476**	0.0160*
		(0.0036)	(0.0034)		(0.0039)	(0.0053)
Age	0.0023**	0.0025**	0.0023**	0.0226**	0.0230**	0.0226**
	(0.0002)	(0.0002)	(0.0002)	(0.0005)	(0.0005)	(0.0005)
Marginal R-squared	0.0112	0.0029		0.0219	0.0049	
R-squared	0.4376	0.4325	0.4376	0.4062	0.3981	0.4063
Observations	3,147,000	3,147,000	3,147,000	3,147,000	3,147,000	3,147,000

*** p<0.01, * p<0.05*

Note. Samples include all establishments responding to Economic Censuses between 1992 and 2012, excluding plants with one employee, for which geographic data is available or could be imputed from address records, with NAICS codes corresponding to business services or manufactures. For all establishments, tract-level employment density is computed as the number of employees in all other establishments (total employees minus employees at the establishment) over the square miles in the tract. Miles to CBD calculates distance to MSA CBD centroid for all tracts. All regressions are at the establishment-year level and control for age of establishment, dummy for establishments established before 1976, industry-CBSA-year fixed effects for the establishment's full SIC or NAICS code, MSA, and year, and cubic polynomials for establishment latitude and longitude. All standard errors clustered at the CBSA level.

Table 2. Establishment characteristics gradients, manufactures

	Log sales PW		Log value added PW		Log TFP		Log employment		Log gross margin PW	
Log employment density, tract	0.0286**		0.0274**		0.0037**		0.0304**		0.0257**	
	(0.0015)		(0.0013)		(0.0007)		(0.0042)		(0.0013)	
Log miles to CBD, tract		-0.0030		-0.0033		-0.0027		0.0410**		-0.0069
		(0.0045)		(0.0052)		(0.0016)		(0.0072)		(0.0029)
Age	0.0032**	0.0033**	0.0039**	0.0039**	-0.0007**	-0.0007**	0.0286**	0.0287**	0.0013**	0.0013**
	(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0001)	(0.0001)	(0.0009)	(0.0009)	(0.0003)	(0.0003)
Marginal R-sq	0.003	0.0001	0.0028	0.0002	0.0001	0	0.0028	0	0.0015	0.0001
R-squared	0.5938	0.592	0.5222	0.5203	0.6407	0.6411	0.5004	0.4999	0.5009	0.5038
Observations	756,000	756,000	756,000	756,000	378,000	378,000	756,000	756,000	756,000	756,000

*** p<0.01, * p<0.05*

Note. Samples include all establishments responding to Censuses of Manufactures between 1992 and 2012, excluding plants with one employee, for which geographic data is available or could be imputed from address records, with NAICS codes corresponding manufactures. Sample in columns 5 and 6 include only firms for which TFP can be calculated. For all establishments, tract-level employment density is computed as the number of employees in all other establishments (total employees minus employees at the establishment) over the square miles in the tract. Miles to CBD calculates distance to MSA CBD centroid for all tracts. All regressions are at the establishment-year level and control for age of establishment, dummy for establishments established before 1976, industry-CBSA-year fixed effects for the establishment's full SIC or NAICS code, MSA, and year, and cubic polynomials for establishment latitude and longitude. All standard errors clustered at the CBSA level.

Table 3. Composition effect on entrants at similar tracts

	Density ranked				Distance ranked			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Log construction expenditures, higher ranked	-0.0023*	-0.0025**			-0.0009	0.0027		
	(0.0009)	(0.0009)			(0.0011)	(0.0023)		
Log construction expenditures, lower ranked			0.0020**	0.0014*			-0.0004	0.0003
			(0.0009)	(0.0007)			(0.0010)	(0.0020)
Constant	4.39	4.35	4.34	4.30	4.36	4.30	4.35	4.32
Observations	920,000	920,000	920,000	920,000	920,000	920,000	920,000	920,000
R-squared	0.38	0.38	0.38	0.38	0.38	0.38	0.38	0.38

*** p<0.01, * p<0.05*

Note. Samples include all establishments less than five years old responding to Economic Censuses between 1992 and 2012, excluding plants with one employee, for which geographic data is available or could be imputed from address records, with NAICS codes corresponding manufactures or business services. Construction expenditures are reported dollar values from REITs active in multiple MSAs reporting to the Census of Finance and Insurance. Rankings refer, in columns one through four, to within-MSA ordering of tracts by establishment density in 1992, and in columns five through eight to ordering of tracts according to their centroid distance to the MSA central business district. All regressions are at the establishment-year level and control for age of establishment, dummy for establishments established before 1976, industry-year fixed effect and tract fixed effects, and cubic polynomials for establishment latitude and longitude. All standard errors clustered at the CBSA level.

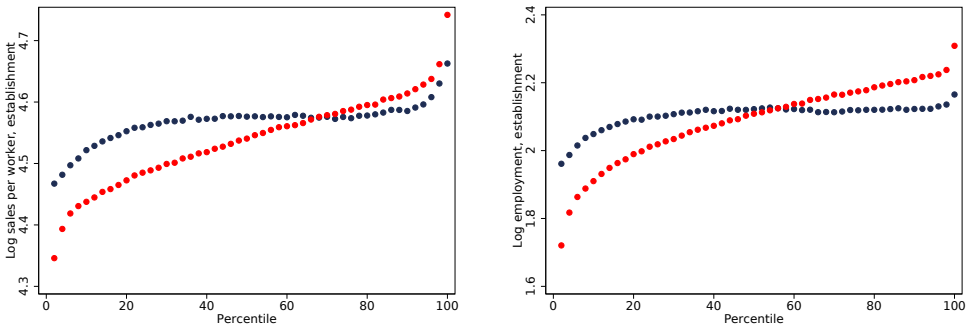
Table 4. Composition effect of construction at other tracts

	Density ranked		Distance ranked		Density ranked	
	(1)	(2)	(3)	(4)	(5)	(6)
	OLS	**OLS**	**OLS**	**OLS**	**IV**	**IV**
Log construction expenditures at all higher-ranked tracts	-0.0064**		-0.0002		-0.0500**	
	(0.0011)		(0.0010)		(0.0057)	
Log construction expenditures at all lower-ranked tracts		0.0043**		0.0012		0.0201**
		(0.0008)		(0.0007)		(0.0031)
Constant	5.31	4.92	4.039	4.02	4.97	4.58
Observations	920,000	920,000	920,000	920,000	920,000	920,000
R-squared	0.30	0.30	0.30	0.30		
Kleiber-Paap F Statistic					62	118

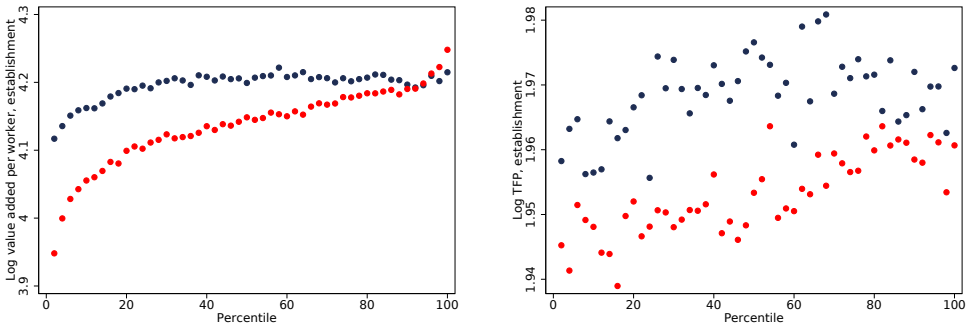
*** p<0.01, * p<0.05*

Note. Samples include all establishments less than five years old responding to Censuses between 1992 and 2012, excluding plants with one employee, for which geographic data is available or imputed, with NAICS codes corresponding manufactures or business services. Construction expenditures are reported dollar values from REITs active in multiple MSAs reporting to the Census of Finance and Insurance, summed over all tracts above the percentile rank of establishment's tract. Rankings refer, in columns one, two, five, and six, to within-MSA ordering of tracts by establishment density in 1992, and in columns three and four to ordering of tracts according to their centroid distance to the MSA central business district. All regressions are at the establishment-year level and control for age of establishment, dummy for establishments established before 1976, industry-year fixed effect and percentile fixed effects, and cubic polynomials for establishment latitude and longitude. All standard errors clustered at the CBSA level.

Figures 1 & 2: Output per worker and employment vs location percentile



Figures 3 & 4: Value added and TFP per worker vs location percentile



Online Appendix

A.1 Proof of sorting equilibrium.

The proof will proceed as follows. First, conditional on there being multiple values of η for different locations, I show that higher η locations must in equilibrium attract higher ψ firms. I then show that differences in η must exist by ruling out cases in which no differences exist in the space S.

A1.1 Sorting conditional on non-degenerate distribution

Assume the opposite, that there exists two locations i and j where i has a higher location specific productivity, so $\eta(i) > \eta(j)$, but firms at i have lower productivity ψ_2 than firms at j, with productivity ψ_1, so $\psi_1 > \psi_2$. Rents must be such that firms in neither location wish to move. Rent is a fixed cost. It is incurred by firms of different productivities in the same way, whereas, by Proposition 2, higher η has differential effects on firms according to their productivity. Lower rent at j must compensate the higher productivity firm at j for decreased variable profit. $\phi(i) - \phi(j) \geq \frac{(\rho\psi_1)^{\sigma-1}}{\sigma}(\eta(i) - \eta(j))$, but since $\psi_1 > \psi_2$, this implies $\phi(i) - \phi(j) > \frac{(\rho\psi_2)^{\sigma-1}}{\sigma}(\eta(i) - \eta(j))$. But this violates the second incentive compatibility constraint of the firm (See equation (4)) at j, $\phi(j) - \phi(i) \geq \frac{(\rho\psi_2)^{\sigma-1}}{\sigma}(\eta(i) - \eta(j))$. Therefore the firm at j could not have optimally chosen j in equilibrium.

A1.2 Proof of impossibility of complete non-sorting equilibrium

Next, I show that an equilibrium in which no location differs by $\eta(i)$, i.e. that $\eta(i)$ is constant for any $i \in S^n$, can be ruled out. The proof evaluates changes in the value of $\eta(i)$ at a convex boundary of the space S. Because at the boundary $\eta(i)$ cannot be constant, non-sorting equilibria can be ruled out.

Again I proceed by contraction, assuming for every $i \in S^n$ $\eta(i) = \eta(\not{i})$. Then we can write the equation for $\eta(i)$, according to equation (10), as

$$\eta(i) = \int_{j \in S} P(i)^{-\sigma}(1 - \xi(i)) \left[\frac{\tau(i,j)}{\rho P(j)}\right]^{1-\sigma} h(j)\psi(j)^{\sigma} \cdot \eta(j)dj.$$

Pulling out the constant value of η and simplifying

$$\text{(A1.1)}\ P(i)^{\sigma} = \int (1 - \xi(i)) \left[\frac{\tau(i,j)}{\rho}\right]^{1-\sigma} P(j)^{1-\sigma} h(j)\psi(j)dj.$$

Differentiating with respect to i and substituting in using equation (12)

$$\text{(A1.2)}\ \sigma P(i)^{-\sigma-1}\nabla_i P(i) \cdot \int_{j \in S} \left[\frac{\tau(i,j)}{\rho P(j)}\right]^{1-\sigma} h(j)\psi(j)dj$$

$$= P(i)^{-\sigma} \cdot (1 - \sigma) \int_{j \in S} \tau(i,j)^{-\sigma} \nabla_i \tau(i,j) P(j)^{\sigma-1} \rho^{\sigma-1} h(j)\psi(j)dj$$

$$\nabla_i P(i) \cdot \int_{j \in S} \sigma P(i)^{-1} \tau(i,j)^{1-\sigma} P(j)^{\sigma-1} \rho^{\sigma-1} h(j)\psi(j)dj$$

$$= (1 - \sigma) \int_{j \in S} \tau(i,j)^{-\sigma} \nabla_i \tau(i,j) P(j)^{\sigma-1} \rho^{\sigma-1} h(j)\psi(j)dj$$

where,

$$\nabla_i P(i) = \int_{j \in S^n} \left[\frac{\tau(i,j)}{\rho}\right]^{1-\sigma} P(j)^{1-\sigma} \psi(j)^{\sigma-1} h(j)dj \frac{\sigma}{1-\sigma} \cdot \int_{j \in S^n} \tau(i,j)^{-\sigma} \nabla_i \tau(i,j) \rho^{\sigma-1} P(j)^{1-\sigma} \psi(j)^{\sigma-1} h(j)dj.$$

Equation (A1.2) can be rewritten as

(A1.3)

$$\int_{j \in S^n} \tau(i,j)^{-\sigma} \nabla_i \tau(i,j) \left(P(j)^{1-\sigma} \psi(j)^{\sigma-1} \cdot z(i) \cdot h(j) + (\sigma - 1) P(j)^{\sigma-1} h(j) \psi(j) \right) dj = 0$$

where

$$z(i) = \int_{j \in S^n} \left[\frac{\tau(i,j)}{\rho} \right]^{1-\sigma} P(j)^{1-\sigma} \psi(j)^{\sigma-1} h(j) dj \frac{\sigma}{1-\sigma} \cdot \frac{}{\int_{j \in S} \sigma P(i)^{-1} \tau(i,j)^{1-\sigma} P(j)^{\sigma-1} \rho^{\sigma-1} h(j) \psi(j) dj}$$

is strictly positive over $i \in S^n$.

Expressing this condition by grouping locations for which $\nabla \tau(i,j)$ is positive, $J_2 \in S^n$, and negative, $J_1 \in S^n$,

$$\int_{J_1} \tau(i,j)^{-\sigma} \nabla_i \tau(i,j) \left(P(j)^{1-\sigma} \psi(j)^{\sigma-1} \cdot z(i) \cdot h(j) + (\sigma - 1) P(j)^{\sigma-1} h(j) \psi(j) \right) dj =$$
$$\int_{J_2} \tau(i,j)^{-\sigma} \nabla_i \tau(i,j) \left(P(j)^{1-\sigma} \psi(j)^{\sigma-1} \cdot z(i) \cdot h(j) + (\sigma - 1) P(j)^{\sigma-1} h(j) \psi(j) \right) dj$$

But in the limit as i approaches the edge of the space, the definition of $\tau(i,j)$ reduces the set of J_2 to zero. While the right hand side of the equation must go to zero, every term on the left hand side is by definition positive, making the sum itself positive. This condition is therefore a contradiction when evaluated at a convex boundary of the space.

A1.3 Partial sorting equilibria

At this point, it may still be possible for some group of locations j to have equivalent location productivity potential $\bar{\eta} = \eta(j)$. In this case, some subset of firms are made indifferent between all locations j, landowners charge $\bar{\phi} = \phi(j)$ constant rents, and density is constant $\bar{h} = h(j)$ across all locations j. Marginal costs of production $w(j)$ may differ across these locations as may the price index $P(j)$ and therefore market access $\frac{\eta(j) P(j)}{w(j)^{1-\sigma}}$.

Although complete non-sorting has been ruled out, this partial sorting equilibrium necessitates a many-to-one match of some set of firm types to locations. In this case, the mapping $\psi(\eta)$ is discontinuous and the rent curve $\phi(\eta)$ is kinked at $\bar{\eta}$.

A2. Equilibrium Conditions Appendix

A2.1 Landowner density and firm choice decisions

A firm's willingness to pay for a space i depends on its variable profit at i vs that at any other location j as well as the rents that firm would face at j. In particular, a firm with productivity parameter ψ will be willing to pay rent for space i according to the function:

$$\text{(A2.1)}\ WTP(\psi, i) = \min_j \left\{\psi^{\sigma-1}\left(\eta(i) - \eta(j)\right) + \phi(j)\right\}$$

where $\phi(j)$ is the rent faced by firms at location j and η is the location-specific productivity parameter. More productive firms derive higher profits from any location, but the rent schedule derived from the IC constraints (equations (3) and (4)) makes it such that firms are unwilling to locate at less productive locations, because the rent savings are outweighed by the productivity loss, and unwilling to locate at more productive locations, because the higher rents must at least offset increased profits.

The IC constraints previously derived ensure rents are such that no firm wishes to move, but this does not immediately imply that the firm with the highest willingness to pay for a particular space is the one assigned to that space. The latter condition would ensure landowners find it optimal to provide density to the firm type matched to their location.

The following argument shows that these IC constraints do ensure just that: assuming a matching function of firms to locations and a schedule of rents supporting the firms incentives to locate, the firm with the highest willingness to pay for a given space is the firm matched to that location.

I first choose to compare the willingness to pay for space i of firm $\psi(\eta(i))$with any firm $\psi < \psi(\eta(i))$.

Choosing specifically the location k to which it is assigned, the willingness to pay of such a firm is less than or equal to the difference between its variable profit at i and k plus rent at k:

(A2.2)
$$WTP(\psi, i) = \min_j \left\{\psi^{\sigma-1}\left(\eta(i) - \eta(j)\right) + \phi(j)\right\} \leq \psi^{\sigma-1}\left(\eta(i) - \eta(k)\right) + \phi(k)$$

However, rearranging the terms in the upper IC constraint for this firm, rent at this location, which is the willingness to pay for i of the firm $\psi(\eta(i))$, or $\phi(i)$ must be greater than the right hand side of the above equation:

$$WTP(\psi, i) = \min_j \left\{\psi^{\sigma-1}\left(\eta(i) - \eta(j)\right) + \phi(j)\right\} \leq \psi^{\sigma-1}\left(\eta(i) - \eta(k)\right) + \phi(k) \leq \phi(i).$$

An identical argument (omitted) is made for firms above ψ, using the lower IC constraint. I therefore conclude that the firm with the highest willingness to pay for location i must be the firm matched to i using the function $\psi(\eta(i))$ in an equilibrium where rents support firm location incentives.

Alternatively, we can write the equilibrium landowner maximization problem

(A2.3) $$\phi(i) = \max_\psi \left\{\min_j \left\{\psi\left(\eta(i) - \eta(j)\right) + \phi(j)\right\}\right\}$$

Again dropping the minimization, i.e. reducing each firm's decision to one between their assigned location and i, and substituting the equilibrium condition for rents (recall)

$$\psi(\eta) = \left[\left(\frac{\partial\phi(\eta)}{\partial\eta}\right)\right]\frac{1}{\sigma - 1}$$

we can rewrite the landowners problem as

$$\text{(A2.4)}\ \phi(i) = \max_{\psi} \left\{ \psi(\eta)^{\sigma-1} \left(\eta(i) - \eta(j)\right) + \int_{\underline{\eta}}^{\eta(j)} \psi(\eta)^{\sigma-1} d\eta \right\}$$

Finally, using the implicit function theorem (since $\eta(j)$ changes with ψ) and a change of variables:

$$\phi(i) = \max_{\psi} \left\{ \psi(\eta)^{\sigma-1} \left(\eta(i) - \eta(\psi(j)\right) + \int_{\psi(\underline{\eta})}^{\psi(\eta(j))} \psi(\eta)^{\sigma-1} \left(\frac{\partial \psi}{\partial \eta}\right)^{-1} d\psi \right\}$$

we can evaluate the first order condition:

$$0 = (\sigma-1)\psi(\eta)^{\sigma-2} \left(\eta(i) - \eta(\psi(j)\right) - \psi(\eta)^{\sigma-1} \left(\frac{\partial \psi}{\partial \eta}\right)^{-1} + \psi(\eta)^{\sigma-1} \left(\frac{\partial \psi}{\partial \eta}\right)^{-1}$$

which is solved where $\eta(\psi(i)) = \eta(\psi(j))$ or by choosing the firm that has already been assigned to that location.

A2.2 General Equilibrium Conditions

Solving the mapping $\psi(\eta)$

Equations (5) through (7) jointly determine, for any ordering η, the set of firms that produce, the arrangement of firms into locations, the matching of productivity to locations, the density of firms at each location, and the rent at each location. Putting equilibrium conditions in equations (5)-(7) together, I find the differential equation governing $\psi(\eta)$:

$$\psi(\eta) = c'' \left(\frac{g(\psi)\psi'(\eta)}{f(\eta)}\right) \cdot \left[\frac{f(\eta)g(\psi)\psi''(\eta) + f(\eta)g'(\psi)\psi'(\eta)^2 - g(\psi)\psi'(\eta)f'(\eta)}{f(\eta)^2}\right]. \tag{8}$$

Note that the boundaries of the location productivity mapping function, $\bar{\eta}$, $\underline{\eta}$, as well as the function $f(\eta)$ are not yet defined.

In the space of η, the four equilibrium conditions provide the solution to an equilibrium where $\eta(i)$ is the mapping of locations to η. It remains to be shown that such an equilibrium exists. To do this, I introduce the last two equilibrium conditions, the worker spatial equilibrium the a balanced trade condition, and use them to solve for $\eta(i)$.

Worker spatial equilibrium condition and wages

Workers must be indifferent between the bundle of goods they can consume at each location. For workers to be indifferent, wages at each location i must exactly offset differences in the price index so workers can attain the same real wage λ across locations.

$$w(i) = P(i) \cdot \lambda$$

Setting wages at a single location j_1 as the numeraire wage, the real wage can be expressed as $\lambda = \frac{1}{P(j_1)}$.

Labor market clearing condition

All labor L must be used in production. For a given firm type ψ, the labor bill is equal to the amount of revenue each produces minus variable profits.

$$l(\eta) = \frac{\sigma \cdot \rho}{\lambda} \cdot \psi(\eta) \cdot \eta$$

Summing over all producing firms and dividing by wages, the total amount of labor must equal the local labor supply, L.

$$L = \int_{\underline{\hat{\psi}}}^{\hat{\psi}_U} \left[\frac{\sigma \cdot \rho}{\lambda} \cdot \psi \cdot \eta(\psi) \cdot g(\psi(\eta)) \right] d\psi(\eta). \tag{9}$$

Where $\underline{\psi}$ is defined by $\underline{\psi} \equiv \phi(\underline{\eta})/\underline{\eta}$.

Balanced trade

Finally, because all three agents consume at their given or equilibrium locations, trade is balanced, i.e. total market for goods at each location must equal the total amount produced at the location. The local demand at j, $R(j)$, is therefore equal to the total revenue of all firms at location j. Recall each firm at location i receive revenue

$$R(j) = P(j) \cdot r(\psi, j) \cdot h(\eta) = P(j) \cdot \psi \cdot \eta(j) \cdot \sigma \cdot h(\eta) \ .$$

Solving for the mapping $\eta(i)$ and the price index.

Equations (5) through (8) characterize any equilibrium in the space of η. While these equations are sufficient to derive predictions. However the general equilibrium requires further mapping of each physical location in S to a value for the index η. I now use the balanced trade condition to solve for this mapping.

To solve for $\eta(i)$, I rewrite the equation for $\eta(i)$ imposing the balanced trade and worker spatial equilibrium conditions together with the mapping of firms to locations $\psi(\eta)$. This yields

$$\eta(i) = P(i)^{-\sigma} \int_{j \in S} \frac{\tau(i,j)^{1-\sigma}}{\rho^{1-\sigma} P(j)^{-\sigma}} \cdot \frac{g(\psi(\eta(j)))\psi'(\eta(j))}{\left(f(\eta)\right)^2} \cdot \left(\psi(\eta(j))\right)^2 \cdot \eta(j) dj \quad (10)$$

while the price index can now be expressed as

$$P(i)^{1-\sigma} = \lambda \cdot \int_{j \in S} \left[\frac{P(j)\tau(i,j)}{\rho\psi(\eta(j))} \right]^{1-\sigma} g(\psi) \cdot \psi'(\eta(j)) \cdot dj \qquad (11)$$

where $\lambda = \dfrac{1}{P(j_1)}$. Equations (9) and (10) constitute a system of nonlinear Hammerstein equations with a kernel of $\tau(i,j)^{1-\sigma}$ the solution to which determines the mappings $\eta(i)$ and $P(i)$.

Finally, note that for any equilibrium mapping of $\eta(i)$, $f(\eta)$ ensures the mass of points in S is accounted for in $[\underline{\eta}, \bar{\eta}]$, such that

$$\int_{\underline{\eta}}^{\bar{\eta}} f(\eta) d\eta = \int_{j \in S} 1 \cdot dj.$$

A3. Proof of existence and conditions for uniqueness

A3.1 Existence in the case of no spillovers

First, equations (10)-(12) can be written as a system of nonlinear Hammerstein equations of the second kind.

Equation (11) in particular can be rewritten as

$$\text{(A3.1)} \quad a(i) = \int \tau(i,j)^{1-\sigma} F\left(a(j), b(j)\right) dj$$

where the functions $a(j) = \dfrac{\eta(j)}{P(j)^{-\sigma}}$ and $b(j) = P(j)^{1-\sigma}$ are defined by an integral equation with a kernel of $\tau(i,j)^{1-\sigma}$ and a nonlinear function

$$F(a(j), b(j)) = \rho^{\sigma-1} \cdot g(\psi(a(j) \cdot b(j)^{\frac{-\sigma}{1-\sigma}})) \psi'(a(j) \cdot b(j)^{\frac{-\sigma}{1-\sigma}}) \cdot \psi(a(j) \cdot b(j)^{\frac{-\sigma}{1-\sigma}})^{2-2\sigma} \cdot a(j) \cdot b(j)^{\frac{1}{\sigma-1}},$$

In turn, the function $b(j)$ can be used to similarly rewrite equation (12) as

$$\text{(A3.2) } b(i) = \int \tau(i,j)^{1-\sigma} G\left(a(j), b(j)\right) dj$$

where the function $b(j)$ is defined over an integral equation with a kernel of $\tau(i,j)^{1-\sigma}$ and a nonlinear function

$$G\left(a(j), b(j)\right) = b(j) \cdot \rho^{\sigma-1} \psi(a(j) \cdot b(j)^{\frac{-\sigma}{1-\sigma}})^{\sigma-1} g(\psi) \cdot \psi'(a(j) \cdot b(j)^{\frac{-\sigma}{1-\sigma}})$$

Together, equations (A3.1) and (A3.2) constitute a system of nonlinear Hammerstein equations of the second kind. Agarwal et al. (2008)show the existence for such a system on four conditions they refer to as (C1)-(C4). I will show that conditions (C1)-(C4) are satisfied in the system described by (A3.1) and (A3.2), and therefore that an equilibrium exists.

First, by assumption, the kernel of this system of equations $\tau(i,j)^{1-\sigma}$ is continuous and non-negative. This satisfies condition (C1). Note that $F\left(a(j), b(j)\right)$ and $G\left(a(j), b(j)\right)$ accept only non-negative arguments and must be non-negative everywhere, and are continuous and closed for any non-negative elements $a(j),\ b(j)$, since, by equation (10), $\psi(\eta(i))$ and $\psi'(\eta(i))$ are continuous functions. This satisfies condition (C2). The distribution of firm productivities is bounded by some maximum $\bar{\psi}$ and $g(\bar{\psi})$, maximum density for some firm type, is finite. Notice that because the rent at each location is finite, the derivative function $\psi'(\eta)$ must be finite everywhere and we can define $\bar{\psi}'$ as its maximum.

Next, define

$$d(i) = \sup \textstyle\int_S \tau(i,j) dj$$

and

$$w_{F1}(a(j)) = a(j),\ w_{F2}(b(j)) = b(j)^{\frac{1}{\sigma-1}},\ w_{G1}(a(j)) = 1,\ w_{G2}(b(j)) = b(j)\ q = \rho^{\sigma-1} \cdot \bar{\psi} \cdot \bar{\psi}' \cdot g(\bar{\psi})$$

Then, by construction, $G\left(a(j), b(j)\right)$ and $F\left(a(j), b(j)\right)$ are both less than $q \cdot w_{F1}(a(j)) \cdot w_{F2}(b(j))$ and $q \cdot w_{G1}(a(j)) \cdot w_{G2}(b(j))$, respectively, which are finite. This satisfies conditions (C3).

Finally, note that

$$\alpha > dq\alpha \frac{2\sigma - 1}{\sigma - 1}$$

for

$$\alpha < (dq) \frac{1 - \sigma}{2\sigma - 1},$$

fulfilling (C4). This completing the sufficient conditions for existence of a nontrivial solution to the system.

A3.2 Uniqueness

In the general case, the equilibrium of this model will not be unique with respect to the mapping of firms to locations. However, two forms of restrictions on the kernel, or the trade costs, and therefore restrictions on the geography underlying the model, admit a single equilibrium.

?shows that the system of equations in (A3.1) and (A3.2) has a unique solution if the following Lipschitz condition is satisfied:

$$\text{(A3.3)} \ \left(F(a_1, b_1) - F(a_2, b_2)\right)^2 \leq k_1^2 \left(\left((a_1 - a_2\right)^2 + \left((b_1 - b_2\right)^2\right),$$

$$\left(G(a_1, b_1) - G(a_2, b_2)\right)^2 \leq k_2^2 \left(\left((a_1 - a_2\right)^2 + \left((b_1 - b_2\right)^2\right),$$

for any possible $a_1,\ b_1,\ a_2,\ b_2$, and some $k_1,\ k_2$ such that

$$k_1^2 + k_2^2 < \lambda$$

where λ is the smallest eigenvector from the kernel defined by

$$K(i, j) = \int_{S^n} \left(\tau(i, r)\tau(j, r)\right)^{1-\sigma} dr$$

A3.3 Stability

The equilibrium is point-wise locally stable if no small group of entrepreneurs or workers can increase their welfare by moving to a different location and no group of landowners can increase profits by changing the amount of density they provide. I show that any equilibrium with a one-to-one, continuous matching $\psi(\eta)$ is stable.

First, no group ϵ of landowners $i \in \epsilon$ can improve profits by adjusting the density of firms at their locations by some fixed amount γ. Adjusting density downward lead (by Appendix 2) to a decrease in profits and to a decrease in $\eta(i)$. Adjusting density upwards for each landowner reduces profits by a fixed amount through increased marginal cost for each landowner, while potentially increasing $\eta(i)$ via higher local demand. As ϵ becomes small, the effect of local changes due to $\epsilon\gamma$ on $\eta(i)$ go to zero, since $\eta(i)$ is defined with respect to all points $j \in S^n$. For any change γ in density, there is an ϵ small enough such that $\triangle\eta(i) < c(h(i) + \gamma) - c(h(i))$. Since this is true for any change in density γ, no arbitrarily small group of landowners can increase profits by deviating in their density provision.

Next, no small group of firms and workers can improve their profits by moving to another location. Firms moving to a new location pay higher rents for the increase in density but may benefit from better market access, as higher demand for their goods from other firms and workers in their deviating group drive up variable profits at the new location. Following Allen and Arkolakis (2013), this cannot be the case when $\frac{d\pi(i)}{dh(i)} < 0$ which is the case if $\frac{d\eta(i)}{dh(i)} < -\frac{d\phi(i)}{dh(i)}$. Intuitively, because $\eta(i)$ is defined with respect to the entire space, smaller groups of firms have an increasingly smaller effect on $\eta(i)$, while the negative effect of density on profits via the direct impact on $\phi(i)$ is constant.

Worker's real wages are always equalized across locations, so no independent move by workers can improve their utility.

A3.4 Incorporating productivity spillovers

The framework laid out in Section 2 can be expanded to incorporate a flexible form of productivity spillovers. In this section I present a model where locations differ by an endogenous location productivity amenity. The index strategy combined with functional form assumptions regarding the relationship between exogenous and endogenous firm productivity allow productivity spillovers to be incorporated into location-specific productivity such that the equilibrium conditions of Section 2 hold with only slight modifications.

As before, firms sell differentiated goods at a markup over marginal cost to all locations $j \in S^n$. However, firm productivity, now denoted by $\tilde{\psi}$ is now endogenously defined by the firm's location. Firm variable profit at location i can be expressed as

$$r(\psi, i)/\sigma = \int_{S^n} \frac{P(i)^{-\sigma} \cdot (1 + \xi(i))^{1-\sigma} \cdot \tau(i,j)^{1-\sigma}}{P(j)^{1-\sigma} \rho^{1-\sigma} \psi(i)^{1-\sigma}} \frac{R(j)}{\sigma} dj$$

where firm productivity is a function of exogenous firm productivity ψ and location-specific productivity spillovers

$$\psi(i) = f(\psi, s(i))$$

The location-specific productive amenities s is a function of the density, productivity, and distance of other firms.

$$s(i) = f(H, \Psi, D)$$

where D is a the (exogenous) distance function between locations, Ψ is the (endogenous) mapping of firm productivities to (all) locations, and H is the (endogenous) function governing densities at all points $j \in S$.

A sufficient condition for isomorphism between this model and the model in Section 2 is for firm productivity and location productivity spillovers to be multiplicatively separable:

$$\psi(i) = \psi \cdot s(i).$$

Note that this model conforms to the standard agglomeration model when $\psi = 1$ for every firm. When both ψ and $s(i)$ are variable, more productive firms will experience larger effects from the same value of $s(i)$, which is an feature of other models in the literature (Gaubert 2013) and suggested by empirical evidence (Combs et al 2012).

Under the above assumption, firm variable profits at i can be expressed as

$$r(\psi, i)/\sigma = \psi \cdot \tilde{\eta}(i)$$

where $\tilde{\eta(i)} = s(i) \cdot \eta(i)$. Landowner, firm, and worker optimal decisions follow as before, now as a function of $\tilde{\eta}$ rather than η.

Note that $s(i)$ is determined endogenously. Because η is endogenous, this does not affect equations (7)-(10). However, the mapping of location productivity $\tilde{\eta}(i)$ to locations is now altered:

$$\tilde{\eta}(i) = s(i) \int_{j \in S} P(i)^{-\sigma}(1 - \xi(i)) \left[\frac{\tau(i,j)}{\rho P(j)}\right]^{1-\sigma} h(j)\psi(j)^{\sigma} \cdot \eta(j)dj.$$

In order to solve for an equilibrium, a solution must be provided for $\tilde{\eta}$ as well as $s(i)$ in conjunction with the unaltered equation for the price index. Following the proof in A3.1, a non-trivial solution to a system of three integral equations of the second kind, provided a specific functional form for $s(i)$ is determined, can be shown to exist. As with $\eta(i)$, restrictions on the value of spillovers will be necessary in order to ensure conditions (C1)-(C4) are met.

A4. Predictions Appendix

A4.1 Gradients

The establishment density-productivity relationship

Using the equilibrium conditions derived in Section 2, I show the model's primary prediction: the productivity-density relationship. Unlike previous theories of firm sorting, this is a prediction, not an assumption of the model. The relationship arises because the most advantageous locations simultaneously attract the most productive firms and command the highest prices, inducing landowners to allow higher establishment density. Both landowners and firms react to the location's characteristic, and the establishment density-productivity relationship arises out of their correlated decisions.

To derive this relationship, I decompose the relationship between firm productivity and density into the relationship between productivity and location advantage, and density and location advantage.

$$\frac{d\psi(\eta)}{dh(\eta)} = \frac{d\psi(\eta)}{d\eta} \cdot \left(\frac{dh(\eta)}{d\eta}\right)^{-1}.$$

$Prediction\ 1$: Locations with higher firm density have more productive firms: $d\psi(\eta)/dh(\eta) > 0$.

The model's first prediction immediately follows from the above equation and Lemma 3.

The establishment density-employment relationship

The model predicts a positive relationship between establishment density and the size of firms. Intuitively, higher demand increases firm size through consumer substitution in two ways. First, more productive firms charge lower prices and capture larger shares of demand, irrespective of

their location. Second, holding firm productivity constant, higher η locations increase demand through a combination of lower local variable costs, and lower transportation costs to relatively more markets (higher local demand). Because firms sort, more productive firms grow larger due to a Matthew effect: the most productive firms, larger in their own right, purchase locations that push them to grow even more.[53]

Recall that the labor force hired by a firm at η can be expressed as

$$l(\eta) = \frac{\sigma \cdot \rho}{\lambda} \cdot \psi(\eta) \cdot \eta.$$

Using the method set out in Section 3.1.1, the above equation leads to the second prediction:

> $Prediction\ 2$: Firms in denser locations will be larger.
>
> Prediction 2 follows from the first order condition in the above equation and the sign of the relationships in Lemma 3. Intuitively, higher η locations have both larger firms and higher firm density, and both effects positively affect the relationship between productivity and employment density.

The employment density-productivity relationship (population density/productivity relationship)

In the model, total employment density at a given location is equivalent to total population density at that location, as there is no commuting.

Prediction 3 follows from the above equation for employment density and Lemma 3.

> $Prediction\ 3a$: Ex-ante more productive firms locate in higher employment and higher employment density areas. $d\psi(\eta)/dd(\eta) > 0$

Intuitively, employment density is simply the firm size at a location times the density of such firms. This relationship would be positive even without

[53] This in turn implies that part of the well-known productivity-scale relationship is a result of location-specific effects.

the density response built into the model; if all landowners had a constant, inelastic supply of space, making the number of firms at each location constant, the fact that firms in more advantageous locations are larger ensures that firm productivity increases with employment density. Here, the additional density response only increases the size of the employment-location advantage elasticity.

However, observed productivity need not necessarily be positively correlated with η. In particular, when productivity spillovers s are large relative to underlying differences in productivity ψ, but have relatively small impacts on profits when compared to differences in market access, firms sort primarily based on market access. Ex-ante more-productive establishments select into locations with fewer productivity spillovers but higher market access, and ex-ante less-productive firms become, ex-post, more productive. Prediction 3b formalizes this intuition.

> $Prediction\ 3b$: The observed relationship between productivity and establishment density, $\epsilon_{TFP/H}$, is positive when (1) sorting is large enough (2) Market access forces are weak enough or (3) productivity spillovers are weak enough.

Note that the observed productivity can be written as $\epsilon_{TFP/H} = \left(\epsilon_{\psi\eta} + \epsilon_{s(i)/\eta}\right) \cdot \epsilon_{h/\eta}^{-1}$. Since $\epsilon_{h/\eta} > 0,\ \epsilon_{TFP/H} > 0\ iff$ ...$\epsilon_{\psi\eta} > -\epsilon_{s(i)/\eta}$. Since $\epsilon_{\psi/\eta} > 0$... $\epsilon_{TFP/H} > 0\ if\ \epsilon_{s(i)/\eta} \geq 0$ or $\epsilon_{TFP/H} > 0\ if\ \dfrac{\epsilon_{\psi/\eta}}{-\epsilon_{s(i)/\eta}} > 1$

but as $\epsilon_{s(i)/\eta}$ is very negative, $\epsilon_{MA/\eta}$ must grow large and positive.

A negative sign could only occur when $s(i)$ is negatively related, on average, to η, which can only occur if firms gain a lot from market access, so much as to offset the productivity they would have gained at lower η locations. At the same time, the differences in endogenous productivity $s(i)$ have to be large enough to offset differences in exogenous productivity $\psi(\eta)$.

Productivity and rent

In the model, firms pay for the advantage of higher demand inherent in higher-η locations. In geography models with pecuniary externalities and

homogenous firms, price differences would arise between locations without firm sorting, but productivity would be constant across locations. The introduction of heterogeneous productivity amplifies price differences and also ensures that more productive firms location in higher priced areas.

The relationship between rents and firm productivity can be expressed as

$$\frac{d\psi(\eta)}{d\phi(\eta)} = \psi(\eta) \cdot \frac{d\psi(\eta)}{d\eta}.$$

Intuitively, firms in locations that are more advantageous are more productive and locations that are more advantageous are more expensive. Prediction 4 immediately follows from the above equation and Lemma 3.

$Prediction\ 4$: More productive firms pay higher rents: $d\psi(\eta)/d\phi(\eta) > 0$.

The density-profitability relationship

While more productive firms pay higher fixed costs to operate in more productive locations, their overall profits are higher. Intuitively, for a location i with a given η, more productive firms have higher sales at i, yet face the same fixed costs (and variable costs). Firms that are more productive than the firm assigned in equilibrium to i would therefore have higher profits at i, yet because of the incentive compatibility constraints, their own location must be more profitable for them. Thus even though fixed costs are higher at high-η locations, the more productive firms that locate there do so precisely because those higher fixed costs are not out-weighed by the variable profit gains they make, and their overall profits remain higher than their less productive counterparts at less costly locations. Finally, these locations are also denser, as higher prices induce landowners to increase the amount of density they provide.

The model yields the following, final major static prediction:

$Prediction\ 5$: Firms in denser locations are more profitable. $d\pi(\eta)/dh(\eta) > 0$.

$Proof$: First, note that $\pi'_f(\eta) > 0$, which is true by taking the first order condition of π_f and substituting equation (4). With Lemma 3, this guarantees the result in $Prediction\ 5$.

A4.2 Composition effects

Prediction 6:

I define the average productivity between some cutoff η_c and the most advantageous location $\bar{\eta}$

$$\tilde{\psi} = \frac{\int_{\psi(\eta_c)}^{\psi(\bar{\eta})} \psi g(\psi(\eta)) d\eta}{\int_{\psi(\eta_c)}^{\psi(\bar{\eta})} g(\psi(\eta)) d\eta}.$$

Equation (6), the real estate market clearing condition, ensures that the total density of firms between productivity levels $\psi(\bar{\eta})$ and $\psi(\eta_c)$ is accounted for in the density of firms present between η and η_c.

$$\int_{\psi(\eta_c)}^{\psi(\bar{\eta})} g(\psi(\eta)) d\psi = \int_{\eta_c}^{\bar{\eta}} h(\eta) f(\eta) d\eta.$$

The real estate market clearing condition drives the composition effect. Because more real estate exists, given a positive shock, within the urban core, more firms must enter. For the condition to hold, the lower bound must move down so that the full mass of firm increases.

To model a shock to the supply density, for some location $\eta_1 \in [\eta_c, \bar{\eta}]$ I assume an idiosyncratic cost of development at that location, introducing a new parameter $\kappa \leq 1$ which affects construction costs for all locations $\eta \in [\eta_1, \bar{\eta}]$. Formerly, construction costs everywhere were identically defined according to the function $c(h(\eta))$. Now, the construction costs are redefined as

$$c_{new}(h(\eta)) \equiv c(h(\eta)) + \kappa(h) \cdot \epsilon.$$

with ϵ arbitrarily small and for a function $\kappa(\eta)$ defined as

$$\kappa(\eta) = \begin{cases} 0 & \eta < \eta_1 \\ a[h(\eta) - h(\eta_1)] & \eta \geq \eta_1 \end{cases}.$$

for some a. This functional form hypothesizes an arbitrarily small shock to density while preserving the smoothness of the functional gradients.

The cost of providing density above η_1 deviates from the otherwise symmetric cost across the rest of the space. A negative a, by equation (5) has a positive effect on density at i. Furthermore, the real estate market clearing condition now becomes

$$\int_{\psi_{new}(\eta_c)}^{\psi(\bar{\eta})} g(\psi(\eta))d\psi = \int_{\eta_c}^{\bar{\eta}} h_{new}(\eta)d\eta.$$

where $h_{new}(\eta)$ is the new density function and $\psi_{new}(\eta_c)$ is the new cutoff firm productivity. In particular $h_{new}(\eta) > h(\eta)$ when $a < 0$.

To accommodate the new density, the left-hand side of the condition must also increase, which is to say the total mass of firms between $\bar{\eta}$ and η_c must expand. But this can only happen by lowering the lower bound, $\psi_{new}(\eta_c) < \psi(\eta_c)$. In order to accommodate more firms in the same space, new, less productive firms that were previously priced out must enter. Prediction 6 follows.

As the sorting pattern changes so that less productive firms enter at each location, the quality of firms at those locations decreases marginally, reducing the density provided to them by the landowners. This attenuates the initial shock, however it cannot reverse the direction.

Prediction 7:

Average firm productivity for this neighboring set of spaces is defined as

$$\tilde{\psi}_{NL} = \frac{\int_{\psi(\eta_{c0})}^{\psi(\eta_{c1})} \psi g(\psi(\eta))d\eta}{\int_{\psi(\eta_{c0})}^{\psi(\eta_{c1})} g(\psi(\eta))d\eta}.$$

If there is any adjustment on the lower margin of firms in $[\eta_{c1}, \eta_{c2}]$, $d\psi(\eta_{c1})/da \geq 0$. But this lower cutoff is also the upper cutoff of the neighboring subset $[\eta_{c0}, \eta_{c1}]$. In addition, shift in the matching function between η_{c0} and η_{c1}

implies a weakly decreasing lower bound, $d\psi(\eta_{c0})/d\kappa \geq 0$, as even after adjustments in density provision in $[\eta_{c0}, \eta_{c1}]$, additional firms enter from below to compensate for firms leaving to $[\eta_{c1}, \eta_{c2}]$. Together, these two margins of adjustment move the average $\tilde{\psi}_N$ in the same downward direction.

A5. Conditions for a mono-centric equilibrium

Urban models and some parts of the new economic geography literature (Fujita Krugman Venables 1999) have traditionally assumed a single, central business district. No such organization of space is assumed in this model. Although the model may yield a single center of economic activity, other equilibria may have multiple "centers". Because cities often have a single business center and because the literature has so often assumed geographies with exogenous centers, in this appendix I examine conditions under which economic activity must necessarily be mono-centric in the space.

The space S is mono-centric if there are no troughs of economic activity. As shown in sections 2 and 3 above, $\eta(i)$ is a sufficient statistic for economic activity at any location. So conditions that exclude local minima in $\eta(i)$ are sufficient conditions to guarantee that any equilibria is mono-centric. For the remainder of this appendix, I refer to local maxima and minima when I discuss the curvature of the function $\eta(i)$ in the space S.

To exclude local minima in $\eta(i)$ is to exclude any equilibria for which, for some $i \in S$, $\nabla\eta(i) = 0$ and $\nabla^2\eta(i) < 0$. To simplify the computation we rewrite

$$\text{(A4.1)}\quad \eta(i) = P(i)^{-\sigma}a(i)$$

where $a(i) = \int_{j\in S} \tau(i,j)^{1-\sigma} z(j)dj$ and $P(i) = \int_{j\in S} \tau(i,j)^{1-\sigma} y(j)dj$.
The first order condition is therefore

$$\text{(A4.2)}\quad \frac{\partial\left(P(i)^{-\sigma}\right)}{\partial i} = -\frac{\partial\left(a(i)\right)}{\partial i}\frac{P(i)^{-\sigma}}{a(i)}.$$

Next, I evaluate the second derivative of $\eta(i)$. At a local minima, this must be positive.

$$\text{(A4.3)}\ \frac{\partial^2 \eta}{\partial i^2} = \frac{\partial^2 \left(P(i)^{-\sigma}\right)}{\partial i^2} a(i) + 2\frac{\partial \left(P(i)^{-\sigma}\right)}{\partial i}\frac{\partial \left(a(i)\right)}{\partial i} + \frac{\partial^2 \left(a(i)\right)}{\partial i^2} P(i)^{-\sigma}$$

With further substitution of the first order condition, equation (A4.3) can be rewritten written as

$$\text{(A4.4)}\ \frac{\partial^2 \eta}{\partial i^2} = \frac{\partial^2 \left(P(i)^{-\sigma}\right)}{\partial i^2} a(i) - 2\frac{P(i)^{-\sigma}}{a(i)}\left(\frac{\partial \left(a(i)\right)}{\partial i}\right)^2 + \frac{\partial^2 \left(a(i)\right)}{\partial i^2} P(i)^{-\sigma}$$

or

$$\frac{\partial^2 \eta}{\partial i^2} = (1-\sigma)\int_{j\in S}\left((-\sigma)\tau(i,j)^{-\sigma-1}\nabla\tau(i,j) + \tau(i,j)^{-\sigma}\nabla^2\tau(i,j)\right)y(j)dj \cdot a(i) - 2\frac{P(i)^{-\sigma}}{a(i)}\left(\frac{\partial \left(a(i)\right)}{\partial i}\right)^2$$

$$+(1-\sigma)\int_{j\in S}\left((-\sigma)\tau(i,j)^{-\sigma-1}\nabla\tau(i,j) + \tau(i,j)^{-\sigma}\nabla^2\tau(i,j)\right)z(j)dj \cdot P(i)^{-\sigma}$$

which is greater than zero if

$$-\sigma\tau(i,j)^{-\sigma-1}\nabla\tau(i,j) + \tau(i,j)^{-\sigma}\nabla^2\tau(i,j) > 0$$

or

$$\text{(A4.5)}\ \nabla^2\tau(i,j) > \sigma\frac{\nabla\tau(i,j)}{\tau(i,j)}$$

When transportation costs are sufficiently convex relative to the elasticity of substitution, no local minima are possible. Intuitively, space in local minima is worse than spaces in any direction. Moving in any direction brings firms closer to their own local demand but further from centers of demand

on the other side of the local minima. Convex transportation costs and substitutability of goods jointly make the such tradeoffs of proximate markets for further markets too dear. The result is that distributions with local centers of activity and local valleys cannot be supported.

It should be noted that this is a sufficient but not a necessary condition for single-peaked equilibria. If this condition is violated, the first term is negative while the second is positive. Depending on the functions $z(j)$, $y(j)$, $a(j)$, and $P(i)$, the second derivative may still be negative everywhere, in some places, or nowhere.

Using the flexible functional form for transportation costs $\tau(i,j) = (1 + \|i - j\|)^d$, where transport costs depend on the distance between points i and j and the parameter d, which reflects the extent of convexity in the transportation cost, the condition in (A4.5) is

$$d > \frac{1 + \sqrt{1 + 4\sigma}}{2}$$

A6. Data Appendix

In sections 5 and 6, I use restricted access US Census Bureau data on all US establishments between 1992-2012. This includes yearly administrative data on employment and payroll from the Census' Longitudinal Business Database (LBD) and US Economic Census data from all economic Censuses from the years 1992, 1997, 2002, 2007, and 2012. In addition, I use IRS establishment-level data to supplement yearly sales data in the LBD and for micro-geographic data at the establishment level, including establishment address, zip code, Census Block and Census Tract. Of all establishments in the LBD in each year – numbering between 6.5 and 7.5 million – I am able to assign tract information to roughly 90%.[54] My sample excludes single-employee establishments. Reported sales in retail, wholesale,

[54]Tract information is provided for between 30-60% of observations each year. Using address matching across multiple observations, I am able to assign an additional tract infor-

and non-tradable service sectors may be reflect differences in local price indexes. To mitigate this issue, I restrict my sample only to tradable sectors. My final sample is composed of about three quarters of a million establishments per year. Appendix Table A.1 reports summary statistics.

The majority of my analysis will be at the tract level. I supplement this basic sample with public Decennial Population Census data on tract population, housing, and demographics, and more detailed data on a subset of establishments from Census Surveys, including the Annual Survey of Manufactures in order to confirm the robustness of my results.

Employment and firm density are constructed using Economic Censuses and SSEL yearly count of active establishments over square miles of land in a tract, respectively. To avoid reflexivity problems, employment density is calculated for each establishment using the employment only at all other establishments in the Census tract (leaving out the establishment's own employment). I construct output per worker as a measure of firm productivity. While output per worker is a measure of productivity commensurate with the model, it is realistic to assume the measure in the data is affected by capital levels and worker heterogeneity. To ensure my results are not driven by these forces that are unaccounted for in my model, I use value added per worker and gross margin (value added minus payroll), available for establishments responding to Census of Manufacturing, as well as total factor productivity, as calculated by Petrin, Reiter, and White (2012), where possible to confirm results. Columns two and three of Appendix Table 1 report summary statistics for these sub-samples, respectively.

Non-residential real estate data is taken from responses to rents at the establishment level from the Censuses of Manufacturers. Although rent does not vary by firm output or employment in the model, I compute rent per worker as an additional metric to ensure plant-size differences are not driving my results.

mation to an additional 40% of firms. For 20%, I use zip codes to impute tract. A remaining 10% of establishments cannot be traced to specific tracts using the data available. A CES paper details the imputation process.

A7 Instrumental Variable Approach

In order to isolate exogenous, marginal changes to local construction costs, I use the inter-city linkages of real estate developers. Commercial real estate development is a leveraged industry (Gyourko, 2009). Developers such as real estate investment trusts, or REITs that also hold an lease real assets may be exposed to real estate shocks via effects on income from these assets. Changes in local real estate prices may therefore affect firms' propensity to supply space.

If such leaser-developers are active in more than one market, local shocks in one market can affect their ability to supply space in other markets. A firm with projects in Boston and Philadelphia, for example, may delay, sell, or scale-down a Boston project due to a negative shock to the market in Philadelphia that depletes the firms' assets. On the other hand, positive shocks to prices in Philadelphia may divert scarce resources away from Boston projects. I term the former effect an income effect and the latter a price effect.

Using the Census of Finance and Insurance, to which all lessors of real estate respond, I isolate over two hundred firms that operate in my period of observation in multiple cities and undertake new commercial real estate construction in at least one. I predict construction expenditures of a given firm's establishment using the construction expenditures and sales of single-unit leaser-developers operating in linked cities, that is, other cities where the developers have an established presence, weighted by their previous-period payroll in each city. Both the income and price effects appear to be operational.

I use these predicted values as instruments for the level of construction expenditures at the tract and city levels in the following subsection in order to find the effects of these construction expenditures on the quality of entrants. The identifying assumption I make is that the level of sales and investment of single-unit establishments in linked cities affects the quality of entrants in another city only via construction expenditures of the linked

firms.

Several possible channels may violate this exclusions restriction and bias the results. Most importantly, the exclusion restriction may be violated if linked cities are exposed to correlated real estate shocks.[55] Columns five and six of Appendix Table 5 test for endogenous linkages by estimating the relationship between the tract-level aggregate predicted values for linked developers and city-level construction expenditures of non-linked developers. Appendix figures A2 and A3 show a binned scattered residual plot of these relationships. Relationships are positive but insignificant. This suggests any direction of bias in the instrument will likely be positive. While positive results should therefore be interpreted with caution, I will focus on the differential effects across tracts and on negative results.

I instrument for supply shocks to construction using the predicted expenditures, where the first stage estimates

$$log(\hat{c}_{p,t-1}) = \beta_0 + \beta_1 \cdot log(pred_{p,t-1}) + \epsilon_{p,t-1}.$$[56]

[55] If these endogenous links are formed between similar cities, and in particular if linked cities share characteristics that expose them to common price shocks, positive shocks in one market will appear as positive space supply shocks to the linked market. This will positively bias the sales instrument and negatively bias the expenditure instrument. In addition to correlated shocks, developers may specialize in similar types of markets, especially markets with particular similar trends. Similar trends in each market may appear and act as correlated shocks, biasing the instrument as above. Alternatively, developers may enter markets endogenously to hedge against idiosyncratic shocks. If endogenous links formed for hedging will negatively bias the sales instrument.

A third and fourth potential for bias enter as development may respond to income and price effects through other margins. The instrument relies on adjustments on the scope and timing of projects. Leaser-developers may select instead to adjust their selection of projects, choosing to delay the lowest-margin projects, or to alter the quality rather than scope or timing of a project. Lower quality sites or sites with lower-quality construction may attract lower quality firms or produce relatively less productive amenities. The former channel negatively biases results while the latter may be a positive bias.

Of these four channels, only site selection and market hedging negatively biases the sales instrument. Because there appears to be no evidence for market hedging, I only use the sales instrument.

[56] Column one of Appendix Table 3 show the relationship between predicted and actual construction expenditures for tract-level aggregates. Controlling for tract fixed effects in columns, the two are positively correlated and significant. Column two adds city and tract level controls. Columns three and four repeat the exercise using changes.

While the first stage is significant at the tract level, once establishment controls are added, the instrument lacks the power to be run on the second composition effect. When aggregating to the percentile and city levels, the instrument gains power.

Appendix Figures and Tables

Appendix Table 1. Summary Statistics

	All Establishments			Business Services			Manufactures		
	Obs.	Mean	St. Dev.	Obs.	Mean	St. Dev.	Obs.	Mean	St. Dev.
Sales (thousands)	3,147,000	6,848	213,000	2,391,000	3,398	218,000	756,000	17,800	196,000
Employment	3,147,000	30.1204	167.0255	2,391,000	23.2556	145.3009	756,000	51.8393	220.7797
Log sales per worker	3,147,000	4.5348	0.9843	2,391,000	4.4405	0.9996	756,000	4.8328	0.8696
Log value added per worker							756,000	4.2160	0.8111
Log TFP							378,000	1.9518	0.5660
Log gross margin per worker							756,000	3.6008	1.0101
Log establishment density, tract	3,147,000	2.7595	2.1259	2,391,000	2.9392	2.1196	756,000	2.1910	2.0442
Log employment density, tract	3,147,000	7.1942	2.1608	2,391,000	7.3568	2.1553	756,000	6.6798	2.0969
Log distance to CBD, tract	3,147,000	3.2987	2.6680	2,391,000	3.2792	2.6998	756,000	3.3604	2.5639

Note. Samples include all establishments responding to Economic Censuses between 1992 and 2012, excluding plants with one employee, for which geographic data is available or could be imputed from address records, with NAICS codes corresponding to business services or manufactures. For all establishments, tract-level employment density is computed as the number of employees in all other establishments (total employees minus employees at the establishment) over the square miles in the tract. Miles to CBD calculates distance to MSA CBD centroid for all tracts.

Appendix Table 2. Predictions

	Location characteristics	Rent	Est. den	Emp. den
	η	$\phi(\eta)$	$h(\eta)$	$d(\eta)$
Firm Characteristics: η		+	+	+
Ex-ante Productivity $\psi(\eta)$	+	+	+	+
Observed Productivity $\hat{\psi}(\eta)$	Conditional	Conditional	Conditional	Conditional
Size $l(\eta)$	+	+	+	+
Profits $\pi(\eta)$	+	+	+	+
Location Characteristics:				
Prices $P(i)$	None	None	None	None
Market Access $MA(i)$	None	None	None	None
Spillovers $s(i)$	None	None	None	None

Appendix Table 3. Gradients, business services

	Log sales per worker		Log employment	
	(1)	(2)	(3)	(4)
Log employment density, tract	0.0556**		0.0904**	
	(0.0027)		(0.0026)	
Log miles to CBD, tract		-0.0513**		-0.0633**
		(0.0038)		(0.0039)
Age	0.0019**	0.0022**	0.0208**	0.0213**
	(0.0002)	(0.0002)	(0.0004)	(0.0004)
Marginal R-squared	0.0137	0.0038	0.0299	0.0075
R-squared	0.3802	0.374	0.3166	0.3041
Observations	2,391,000	2,391,000	2,391,000	2,391,000

*** p<0.01, * p<0.05*

Note. Samples include all establishments responding to Economic Censuses between 1992 and 2012, excluding plants with one employee, for which geographic data is available or could be imputed from address records, with NAICS codes corresponding to business services. For all establishments, tract-level employment density is computed as the number of employees in all other establishments (total employees minus employees at the establishment) over the square miles in the tract. Miles to CBD calculates distance to MSA CBD centroid for all tracts. All regressions are at the establishment-year level and control for age of establishment, dummy for establishments established before 1976, industry-CBSA-year fixed effects for the establishment's full SIC or NAICS code, MSA, and year, and cubic polynomials for establishment latitude and longitude. All standard errors clustered at the CBSA level.

Appendix Table 4. Factor models

	All	Business services	Manufactures	
	(1)	(2)	(3)	(4)
Establishment variables				
Log sales per worker, establishment	0.4407	0.4325	0.5331	0.8075
Log employment, establishment	0.5938	0.5864	0.7138	0.5939
Log sales	0.7358	0.7354	0.8322	0.8903
Log TFP				0.2425
Tract variables				
Log establishment density	0.7516	0.7706	0.5517	0.1614
Log miles to CBD	-0.4097	-0.4285	-0.2612	-0.0749
Log employment density	0.7635	0.7786	0.5893	0.2056
Log rent per worker	0.2041	0.2037	0.1992	0.1135
Observations	3,147,000	2,391,000	756,000	378,000

Note. Sample in column one includes all establishments responding to Economic Censuses between 1992 and 2012. Sample in column two includes all establishments responding to Censuses of Service between 1992 and 2012 which have NAICS or SIC codes corresponding to business services. .Sample in column three includes all establishments responding to Economic Censuses of Manufactures between 1992 and 2012. Sample in column four includes the subset for which TFP can be calculated. All samples exclude plants with one employee, for which geographic data is available or could be imputed from address records, with NAICS codes corresponding manufactures. For all establishments, tract-level employment density is computed as the number of employees in all other establishments (total employees minus employees at the establishment) over the square miles in the tract. Miles to CBD calculates distance to MSA CBD centroid for all tracts. Establishment density is the establishment count divided by the tract square milage, and rent per worker is the average rent per worker of all reporting establishments in the tract. All variables are at the establishment-year level and residualized for controls including age, dummy for establishments established before 1976, industry-CBSA-year fixed effects for the establishment's full SIC or NAICS code, MSA, and year, and cubic polynomials for establishment latitude and longitude. All standard errors clustered at the CBSA level.

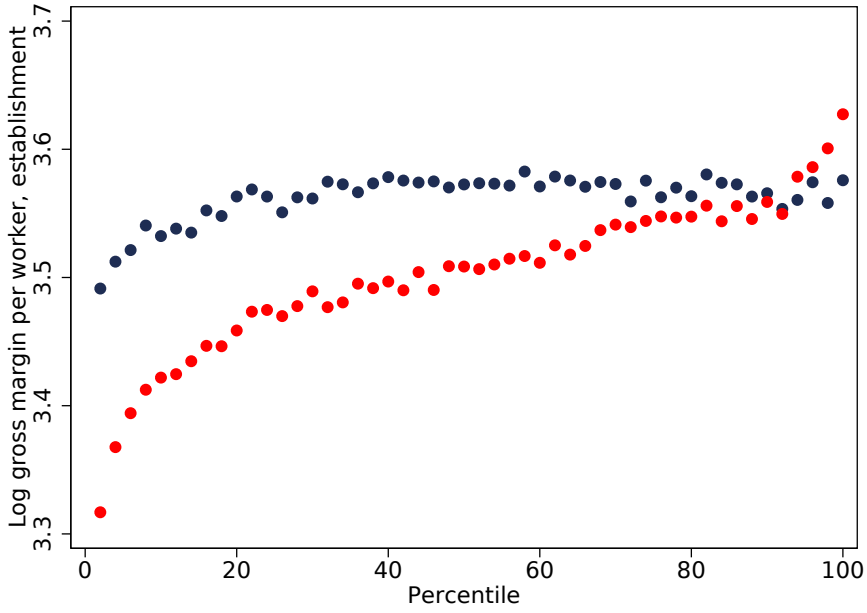
Appendix Table 5. Construction at unlinked firms in linked cities

VARIABLES	**Log construction exp. of unlinked firms, CBSA**			
	(1)	(2)	(3)	(4)
	Levels		Changes	
Log predicted sales, tract	0.009	0.004		
	(0.020)	(0.019)		
Change in log predicted sales, CBSA level			0.004	0.006
			(0.015)	(0.015)
CBSA size	No	Yes	No	Yes
CBSA/Year FE	Yes	Yes	No	No
Constant	1.03	-604	0	-22.25
Observations	4,088	4,088	4,088	4,088
R-squared	0.81	0.84	0.03	0.07

*** p<0.01, ** p<0.05, * p<0.1

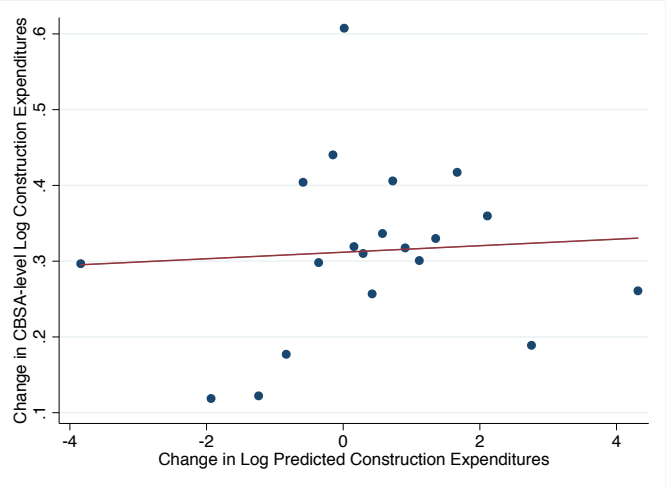
Note. Sample includes all tracts for which Real Estate Investment Trusts respond to questions in the Census of Finance and Insurance respondents on expenditures on new construction on commercial real estate, which have establishments reporting sales at least one other CBSA, and for which geographic information was present or could be imputed from address files. Log construction expenditures at the CBSA level is the log of the sum of the dollar value of all construction expenditure, including office and manufacturing space, reported by all single-unit (i.e., unlinked) REITs in the CBSA. Columns one and two report this measure in levels and columns three and four report changes between census years at the tract level. Log predicted sales are the predictions based on current-year sales of single-unit firms responding to the same census questions in CBSAs where the multi-unit REIT responding in the observed tract has other establishments present, weighted by the percent of sales at the firm level in that CBSA in the previous year. Changes report log differences for this predicted value. All regressions are at the tract level and include controls for year fixed effects. All standard errors clustered at the CBSA level.

Appendix Figure A1: Establishment gross margin vs location percentile[57]



[57]Note. Figure is binned scattered of residuals of regression in Columns 7 and 8 of Table 2. Bins are shown by percentile, with higher density and closer percentiles displayed on the right. Samples include all establishments responding to Economic Censuses between 1992 and 2012, excluding plants with one employee, for which geographic data is available or could be imputed from address records, with NAICS codes corresponding to manufactures. For all establishments, tract-level employment density is computed as the number of employees in all other establishments (total employees minus employees at the establishment) over the square miles in the tract. Miles to CBD calculates distance to MSA CBD centroid for all tracts. All regressions are at the establishment-year level and control for age of establishment, dummy for establishments established before 1976, industry-CBSA-year fixed effects for the establishment's full SIC or NAICS code, MSA, and year, and cubic polynomials for establishment latitude and longitude

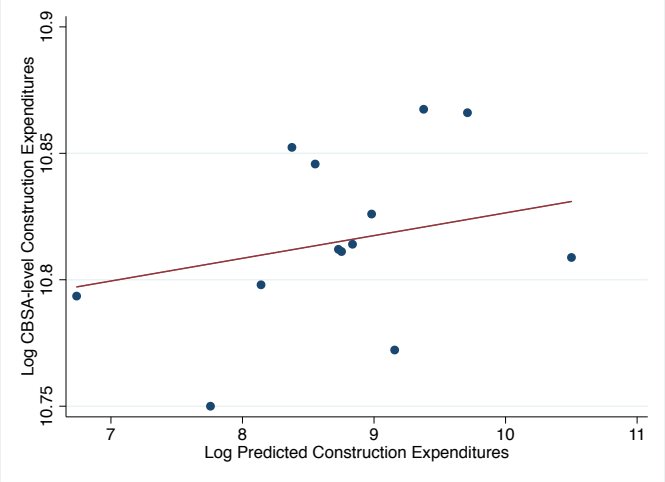
Figure A2: Predicted construction expenditures vs CBSA-level sums, levels



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[58]Note. Figure is binned scattered of residuals of regression in Column 2 of Appendix Table A.5. Sample includes all tracts for which Real Estate Investment Trusts respond to questions in the Census of Finance and Insurance respondents on expenditures on new construction on commercial real estate, which have establishments reporting sales at least one other CBSA, and for which geographic information was present or could be imputed from address files. Log construction expenditures at the CBSA level is the log of the sum of the dollar value of all construction expenditure, including office and manufacturing space, reported by all single-unit (i.e., unlinked) developers in the CBSA. Log predicted sales are the predictions based on current-year sales of single-unit firms responding to the same census questions in CBSAs where the multi-unit developers responding in the observed tract has other establishments present, weighted by the percent of sales at the firm level in that CBSA in the previous year.

Figure A3: Predicted construction expenditures vs CBSA-level sums, changes



[59]

[59]Note. Figure is binned scattered of residuals of regression in Column 4 of Appendix Table 5. Sample includes all tracts for which Real Estate Investment Trusts respond to questions in the Census of Finance and Insurance respondents on expenditures on new construction on commercial real estate, which have establishments reporting sales at least one other CBSA, and for which geographic information was present or could be imputed from address files. Change in log construction expenditures at the CBSA level is the change between census years of the log of the sum of the dollar value of all construction expenditure, including office and manufacturing space, reported by all single-unit (i.e., unlinked) developers in the CBSA. Change in log predicted sales are the predictions based on current and previous census-year sales of single-unit firms responding to the same census questions in CBSAs where the multi-unit developers responding in the observed tract has other establishments present, weighted by the percent of sales at the firm level in that CBSA in the previous year.